

Chapter 6: Investigation Planning

Scope, Evidence, Documentation, and Case Management

Companion Workbook to the Instructional Video

How to Use This Workbook

This is the written companion to the instructional video on investigation planning. It works as a follow-along guide, a standalone textbook chapter, and a practice manual.

Where this chapter sits in the series. Chapter 1 established what intelligence is. Chapter 2 established what OSINT is. Chapter 3 covered the source categories. Chapter 4 covered the collection methodology as a process. Chapter 5 covered the reasoning inside that process. **This chapter covers the administration that holds all of it together.**

That word "administration" undersells it. Planning, documentation, evidence integrity, and case management are what determine whether excellent reasoning survives contact with a reviewer, a lawyer, an auditor, an editor, or a court. An investigation can ask perfect questions, test hypotheses rigorously, and reach a correct conclusion, and still be worthless because nobody can prove where the evidence came from, or because the work took eleven weeks when the decision was in three, or because the case file is a folder of screenshots with no dates.

The labs are chained. All five run on one fictional investigation, from plan through closure. Lab 4 uses a set of simulated digital evidence files provided alongside this workbook, with real hash values, one of which does not match its manifest entry. Do the labs in order and keep every artifact.

Time estimate:

Component	Estimated time
Video	60 to 75 minutes
Reading material	5 to 6 hours
Knowledge assessment	1 hour
Critical thinking exercises	2 hours
Labs 1 through 5 (chained)	12 to 17 hours
Capstone assignment	6 to 9 hours
Total	27 to 36 hours

Materials: a word processor, a spreadsheet application, a file archiver, and a way to compute file hashes. Section 8.4.2 gives the commands for Windows, macOS, and Linux, all built in, so nothing needs to be installed.

The Rules Still Apply

Rule 1: No private individuals. Rule 2: Publicly available means genuinely public. Rule 3: Document as you go. Rule 4: No dark web access. Rule 5: Automation is constrained. Rule 6: A relationship is a claim, and every claim needs a basis.

This chapter adds one more, and it is the chapter's organizing principle:

Rule 7: If it is not written down, it did not happen. This is the oldest rule in investigative work and the one students find hardest to internalize, because it feels like bureaucracy until the first time a finding is challenged. An action you took but did not record cannot be relied on, cannot be reproduced, and cannot be defended. A file you collected but did not log is not evidence, it is a file. Section 1.6 explains why this is a substantive rule about the reliability of your own work rather than a procedural nuisance.

Learning Objectives

Knowledge objectives (you can explain it):

1. Explain what an investigation plan is, what it contains, and what specifically fails without one.
2. Define scope and explain the four dimensions along which it is bounded.
3. Explain scope creep, its causes, and the mechanism that controls it.
4. Distinguish scope, objectives, and priorities, and explain how each constrains the next.
5. Write objectives that are measurable, and explain why an unmeasurable objective cannot be completed.
6. Distinguish strategic goals from tactical tasks and explain how tasks are derived from goals.
7. Describe the prioritization methods available and when each is appropriate.
8. Explain what an evidence collection plan contains beyond a list of sources.
9. Describe the differences between physical and digital evidence and the handling consequences of each.
10. Explain what documentation an investigation must produce and why each artifact exists.
11. Explain chain of custody: what it is, what breaks it, and what a break costs.
12. Describe hashing, verification, and preservation, and explain what each does and does not establish.
13. Explain case management as a discipline and describe the lifecycle of a case from initiation to closure.

Application objectives (you can do it):

1. Write a scope statement with inclusions, exclusions, boundaries, and assumptions.
2. Write measurable objectives with success criteria and completion tests.
3. Prioritize investigative tasks using a stated method and defend the ranking.
4. Build an evidence collection plan covering sources, methods, legal considerations, documentation requirements, and preservation.
5. Maintain investigator notes, evidence logs, screenshot records, timelines, and decision records to a professional standard.
6. Acquire, hash, log, transfer, verify, and preserve digital evidence with an unbroken custody record.
7. Detect and correctly report a chain of custody break or a hash mismatch.
8. Run a case from initiation through closure, including task management, lead tracking, review, and archival.

Analysis objectives (you can judge it):

1. Evaluate an investigation plan and identify what would fail under review.
 2. Assess whether a scope statement actually bounds the work.
 3. Judge whether documentation would allow another investigator to reproduce the work.
 4. Determine whether a custody record is defensible and what a challenge would target.
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Part 1: Reading Material

1.1 Introduction to Investigation Planning

1.1.1 What an Investigation Plan Is

An **investigation plan** is a written document, produced before substantive work begins, that states what the investigation will and will not cover, what it must accomplish, in what order, using what evidence, documented how, managed by whom, and finished when.

It is not a formality and it is not a schedule. It is the artifact that converts a concern into work that can be executed, reviewed, defended, and finished.

What a complete plan contains:

Section	Answers
Authority and initiation	Who authorized this, under what authority, and when

Section	Answers
Background	What prompted it, stated as facts and as a concern
Scope	What is in, what is out, what the boundaries are
Objectives	What must be accomplished, measurably
Priorities	In what order, and what happens if resources fall short
Questions and hypotheses	Carried from Chapter 5
Evidence collection plan	Sources, methods, legality, preservation, documentation
Documentation plan	Which artifacts, in which formats, stored where
Chain of custody procedures	How evidence integrity is established and maintained
Case management	Roles, task tracking, review points, communication, closure criteria
Legal and ethical framework	Applicable constraints, prohibited activities, individuals policy
Risk and constraints	Deadlines, budget, access limits, OPSEC
Success criteria and closure	How we know we are done
Approvals and revision history	Who signed, when, and what changed

1.1.2 What Fails Without a Plan

Four failures, each with a recognizable signature.

Without...	The failure	How it looks in practice
Scope	The investigation expands without limit	Week six, still collecting, no one can say what would finish it
Objectives	Effort without completion	A large amount of work and no answer to "did we accomplish what we set out to?"
Priorities	Effort spent on the wrong things	The interesting thread is fully worked; the decision-relevant one is untouched at the deadline
Documentation and custody	Findings that cannot be defended	The conclusion is correct and collapses under challenge because nobody can show where item 14 came from

A fifth failure sits above all of them: without a plan there is nothing to review. A reviewer cannot tell

whether an investigation was competent, because there is no statement of what it was trying to do. This is why organizations that take investigations seriously require an approved plan before substantive work, and why a plan is worth writing even for a two-day effort.

1.1.3 Planning Is Proportional, Not Optional

The most common objection is that this is too heavy for small work. The answer is the same as Chapter 4's: the plan compresses, and no section disappears.

Plan element	Two-day investigation	Six-month investigation
Authority	One line: who asked, when	Formal tasking memorandum with authority cited
Scope	Three bullets: in, out, boundary	Full scope statement with change control
Objectives	Two measurable objectives	Objective hierarchy with success criteria per objective
Priorities	A numbered list	Weighted matrix with resource allocation and contingency
Evidence plan	A table of six rows	Full plan with legal review per line and preservation procedures
Documentation	A notes file and an evidence log	Documentation standard, templates, storage architecture, retention
Custody	A log with hashes	Full procedures, transfer forms, secure storage, verification schedule
Case management	A task list	Case management system, roles, review cadence, communication plan
Closure	A sentence stating what finishes it	Formal closure criteria, review, archival, retention schedule
Total plan length	One page	Fifteen to forty pages

The one-page plan is a real artifact and it is worth writing. Most investigations that go wrong at small scale go wrong because nobody spent the twenty minutes.

1.1.4 The Planning Flow

TRIGGER
(a concern, a referral, an incident, a request)

AUTHORIZATION
Who can authorize this? Under what authority? Is it recorded?

SCOPE
Subject, time period, geography, activity.
Inclusions, exclusions, boundaries, assumptions.

OBJECTIVES
What must be accomplished. Measurable.
Each with a success criterion.

PRIORITIES
Order and resource allocation.
What gets dropped if we run short.

EVIDENCE
PLAN

DOCUMENTATION
PLAN

CUSTODY
PROCEDURES

CASE MANAGEMENT
Roles, tasks, leads, reviews, communication,
closure criteria

APPROVAL
Signed, dated, version-controlled

EXECUTION

PLAN REVISION (expected, not exceptional)
Any scope or objective change is a
documented amendment, approved, versioned

The revision box is not a footnote. Plans change in nearly every real investigation. What distinguishes a professional effort is that changes are *amendments*, approved and versioned, rather than silent drift.

1.1.5 The Plan Is a Defensive Document

A point students underestimate. When an investigation is challenged, the challenge almost never begins with the conclusion. It begins with the process:

- Why did you investigate this and not that?
- What authorized you to look at this?
- Where did this item come from, and how do you know it is unaltered?
- Who else handled it?
- Why did you stop looking when you did?
- What did you decide not to pursue, and why?

Every one of those questions is answered by a planning artifact, not by analysis. An investigation with brilliant reasoning and no plan cannot answer a single one of them, which is why the plan is the first thing an opposing party asks for and the first thing a competent reviewer reads.

1.2 Investigation Plan Scope

1.2.1 What Scope Is

Scope is the boundary of the investigation. It states what is inside the work and what is outside it, so that everyone (the investigator, the consumer, the reviewer, and any subject) can tell the difference.

Scope is not the same as the question, and it is not the same as the objectives. The question is what you want to know. The scope is where you are permitted and intending to look for it.

1.2.2 The Four Dimensions

Every scope statement bounds the work along four axes. Omitting any one leaves an open edge through which the investigation will expand.

Dimension	Question	Example
Subject	Which entities, systems, accounts, or activities are in scope?	Larkspur Outfitters' brand presence and the sellers listing its products; not Larkspur's employees
Temporal	What period does the investigation cover?	Listings and activity from 1 January of the current year to the present
Geographic / jurisdictional	Which places, jurisdictions, or platforms?	Marketplaces accessible from the United States; the county in which the distribution address sits

Dimension	Question	Example
Activity	What types of activity are being examined?	Unauthorized use of the brand and unauthorized distribution; not pricing practices of authorized dealers

A scope statement that names all four is bounded. One that names two is a suggestion.

1.2.3 Inclusions, Exclusions, Boundaries, and Assumptions

A complete scope section has four parts, and the second and third are the ones that do the work.

Inclusions state what is in. Specific enough to be executed.

Exclusions state what is deliberately out, and why. This is the most valuable and most omitted part of a scope statement. Writing "this investigation does not examine the conduct of Larkspur employees; any indication of internal involvement will be referred rather than pursued under this plan" prevents an entire class of drift and, equally important, tells a later reviewer that the omission was a decision rather than an oversight.

Boundaries state the limits on method and access: what will not be done, which sources will not be used, what requires escalation. The prohibited-activity statement from Chapter 4 lives here.

Assumptions state what the plan takes as given: that a system's logs are retained for 90 days, that a records portal covers the relevant period, that a platform's terms permit manual review. Each assumption is a risk if wrong, and Chapter 5's assumption register begins here rather than at analysis.

1.2.4 A Worked Scope Statement

From the fictional Larkspur Outfitters case used in the labs.

Scope

Inclusions. This investigation covers (a) online listings offering products bearing the Larkspur Outfitters marks on publicly accessible marketplaces and social commerce platforms; (b) the publicly available identity, registration, and infrastructure information associated with the sellers of those listings; (c) publicly available corporate, court, permit, and property records concerning any business entity identified through (b); and (d) the circumstances under which a Larkspur internal product specification document became publicly available, to the extent determinable from public sources.

Temporal boundary. Activity from 1 January of the current year to the date of the final report. Historical corporate and court records may be examined beyond this period where necessary to establish entity identity or history.

Geographic and jurisdictional boundary. Marketplaces and platforms accessible without restriction from the United States. Records research is limited to United States federal and state sources, and to the specific counties identified during collection.

Activity boundary. Unauthorized use of the marks and unauthorized distribution of goods bearing them, and the public availability of the specification document.

Exclusions. This investigation does not examine: (a) the conduct of Larkspur employees or contractors, which if indicated will be referred to Larkspur's general counsel rather than pursued under this plan; (b) authorized dealers' pricing or channel conduct; (c) any individual seller's personal circumstances beyond their identity as a commercial seller; (d) any minor; (e) any matter requiring access to non-public systems or accounts.

Method boundaries. No purchases will be made without written authorization from the client and counsel. No accounts will be created that misrepresent identity. No platform will be accessed by automated means. No communication will be initiated with any seller.

Assumptions. (1) Marketplace listings remain available for at least 30 days, so capture must be prompt. (2) The relevant state corporate registry is current within 30 days. (3) The client's own records of authorized dealers are complete. Each assumption is registered and will be tested.

Notice what the exclusions do. They protect the subjects who are out of scope, they prevent drift, they tell the client what they are not buying, and they give a reviewer the ability to say "the plan says this was deliberate."

1.2.5 Scope Creep

Scope creep is the gradual expansion of an investigation beyond its stated boundary, without a decision to expand it.

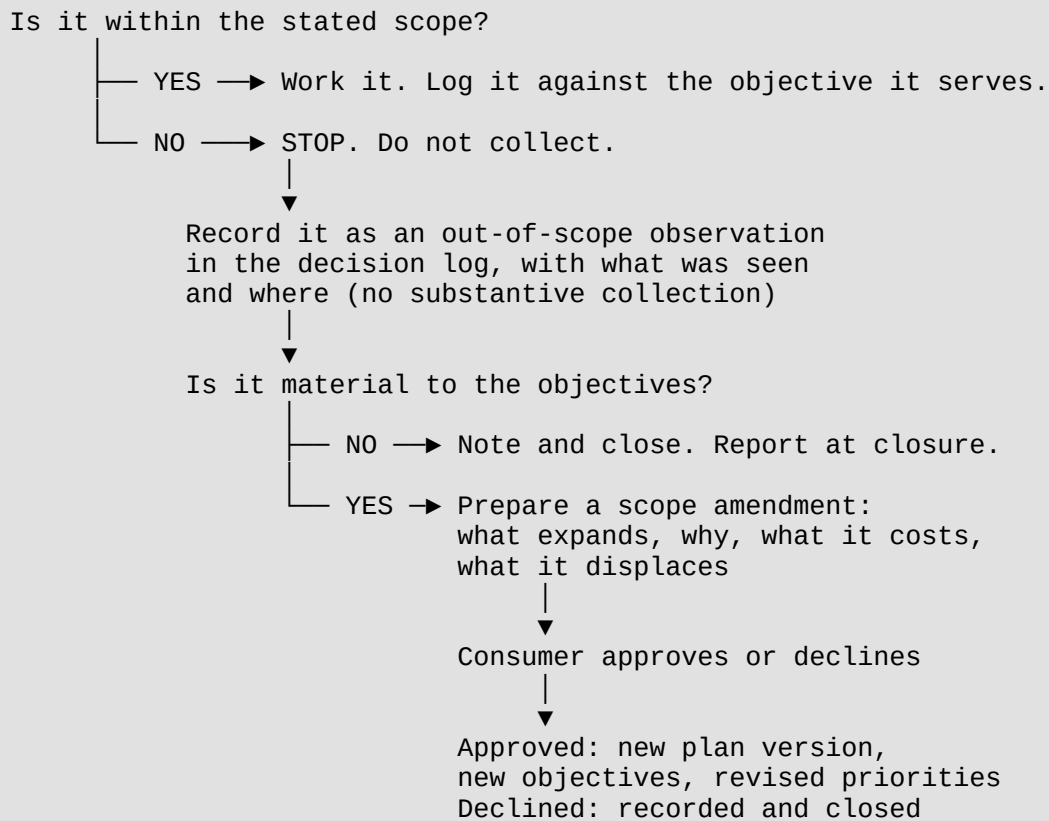
Why it happens:

Cause	Mechanism
An interesting thread appears	Curiosity is not a tasking, and it feels like diligence
A finding suggests a wider problem	Frequently legitimate, and it requires a decision, not a drift
The consumer adds a request informally	"While you're at it, can you also look at..."
An adjacent question is cheap to answer	Cheapness is not relevance
The investigator wants a complete picture	Completeness is not the goal; answering the objectives is

What it costs: the deadline, the budget, the focus on the decision-relevant question, and, in regulated or legally exposed work, the defensibility of the entire effort. An investigator who collected material outside the authorized scope has a problem that the quality of the material does not solve.

The control is a scope change procedure, written into the plan:

NEW THREAD APPEARS



The critical feature of this flow is that the default is stop. An investigator who works an out-of-scope thread first and seeks approval afterward has already created the exposure the procedure exists to prevent.

1.2.6 Scope Statements That Fail

Failing statement	Why it fails
"Investigate the counterfeiting problem"	No dimension bounded; unlimited
"Investigate all activity relating to Larkspur Outfitters"	Subject unbounded; would include the client's own staff
"Determine who is responsible"	This is an objective, not a scope, and it presupposes an actor
"Examine the last several years"	Temporal boundary not specific enough to test completion
"Use all available sources"	No method boundary; invites legal exposure
A scope with inclusions and no exclusions	Every adjacent thread appears permitted

1.3 Investigation Plan Objectives

1.3.1 What an Objective Is

An **objective** is a statement of what the investigation must accomplish, written so that its completion can be verified. Scope says where you may look; objectives say what you must produce.

The test of an objective is simple and unforgiving: **can two people, looking at the work, agree on whether it has been achieved?** If not, it is a wish.

1.3.2 Measurability

An objective is measurable when it names a deliverable, a standard, or a determination that can be checked.

Not measurable	Measurable
"Understand the counterfeiting problem"	"Produce an inventory of all listings offering Larkspur-marked goods on the four in-scope platforms as of the capture date, with each listing captured, hashed, and logged"
"Identify the sellers"	"For each seller account in the inventory, determine whether a business entity can be identified from public records, and record either the identified entity with its supporting records or a documented negative finding with its coverage boundary"
"Look into the document leak"	"Determine the earliest identifiable public appearance of the specification document, and produce a sourced timeline of its subsequent appearances"
"Assess the risk"	"Produce a written assessment of the volume, geographic concentration, and apparent commercial scale of the unauthorized listings, with a confidence level and named gaps"

Note what the measurable versions permit. Each can be completed. Each can be reported as done or not done. Each has a defined output. And each accommodates a negative result: "no entity could be identified, and here is what was searched" is a completed objective, not a failure.

1.3.3 The Objective Record

Each objective gets a record:

Field	Content
Objective ID	Referenced in tasks, evidence, and the report
Statement	One sentence, measurable

Field	Content
Rationale	Why this matters to the decision
Success criterion	What completion looks like, concretely
Completion test	How a reviewer verifies it
Dependencies	Which objectives must complete first
Priority	From Section 1.4
Owner	Who is accountable
Deadline	Tied to the decision
Status	Not started, in progress, complete, blocked, descope
Evidence produced	Which items and artifacts satisfy it

1.3.4 Strategic Goals and Tactical Tasks

An investigation operates at two levels, and confusing them is a common planning failure.

A strategic goal is what the investigation is for. It is usually one sentence, tied to the decision the consumer must make, and it does not change during the investigation.

Objectives sit under the goal. There are usually three to seven.

Tactical tasks sit under objectives. They are the individual units of work, each assignable, each with a duration, each producing a specific output.

STRATEGIC GOAL

"Give Larkspur's general counsel the evidence base needed to decide, by 30 June, whether to pursue platform enforcement, civil action, or referral."

- OBJECTIVE 1: Inventory the listings
 - Task 1.1 Identify in-scope platforms and record the selection basis
 - Task 1.2 Search each platform using the agreed term set
 - Task 1.3 Capture each listing to the documentation standard
 - Task 1.4 Hash and log every capture
 - Task 1.5 Produce the listing inventory
- OBJECTIVE 2: Identify sellers where public records permit
 - Task 2.1 Extract seller identifiers from the inventory
 - Task 2.2 Entity expansion on each candidate name
 - Task 2.3 Registry, court, permit, and property searches
 - Task 2.4 Record identifications and negative findings
- OBJECTIVE 3: Establish the document's public history
- OBJECTIVE 4: Assess scale and concentration

	Strategic goal	Tactical task
Number	One	Dozens
Changes during the investigation	Rarely, and only by amendment	Constantly, as work proceeds
Owner	The consumer, jointly with the lead	An individual investigator
Duration	The whole investigation	Hours to days
Measured by	Whether the decision was supported	Whether the output exists
Written by	The lead, with the consumer	The lead or the investigator
Failure signature	The investigation succeeds and the decision is not supported	A task is complete and nobody can say what objective it served

The most common error at this level is a task list with no objectives above it. The work gets done, everyone is busy, and at the end there is no way to say whether the investigation accomplished anything, because nothing stated what accomplishment would look like.

1.3.5 Success Criteria and the Definition of Done

Every objective needs a **success criterion** (what completion looks like) and a **completion test** (how someone else verifies it).

Objective	Success criterion	Completion test
Inventory the listings	Every listing found on the in-scope platforms during the collection window is captured, hashed, logged, and represented in the inventory with its capture timestamp	A reviewer selects five inventory rows at random and confirms that each has a capture file, a hash, a log entry, and a timestamp
Identify sellers	Each seller account has either an identified entity with supporting records, or a documented negative finding with its coverage boundary	A reviewer confirms no seller row is blank and every negative names the systems searched
Establish document history	The earliest identifiable public appearance is stated with its evidence, and subsequent appearances are on a sourced timeline	A reviewer traces the earliest-appearance claim to a captured, hashed item

Why the completion test matters. Without it, "done" is a claim by the person who did the work. With it, done is a checkable state. In organizations that review investigations, the completion test is what a reviewer actually performs.

1.3.6 Objectives That Fail

- **Unmeasurable.** "Understand," "explore," "look into," "assess the situation" with no defined output.
- **Compound.** Two or three objectives joined into one, so partial completion cannot be reported.
- **Presupposing.** "Document the seller's infringement" assumes the conclusion the investigation is meant to test.
- **Unachievable within scope.** An objective requiring sources the scope excludes.
- **Orphaned.** An objective serving no strategic goal, which usually means it survived from an earlier version of the plan.
- **Without a negative path.** An objective that can only be completed by finding something, which quietly pressures the investigator to find it.

That last one deserves emphasis. **Every objective must be completable by a negative finding.** "Determine whether X can be established from public records" can be completed either way. "Establish X" can only be completed one way, and an investigator working under it is being paid to reach a particular result.

1.4 Investigation Plan Priorities

1.4.1 Why Prioritization Is a Planning Act

Priorities are decided in the plan, not during execution, for one reason: **in the middle of an investigation, everything feels urgent and the loudest thread wins.** A ranking written in advance, with a stated method, is what protects the decision-relevant work from the interesting work.

Prioritization also answers a question every plan must answer and most do not: **what gets dropped if we run short?** If the answer is not written down, the answer in practice is "whatever was last on the list when the deadline arrived," which is not a decision.

1.4.2 Prioritization Methods

Method	How it works	Best for	Weakness
Decision impact	Rank by how much the output would change the consumer's decision	Almost all investigations; the default	Requires knowing the decision precisely
Perishability	Rank by how fast the evidence disappears	Online listings, social content, logs with retention limits, volatile systems	Can crowd out decision-relevant but stable work
Dependency	Rank by what must precede what	Investigations with sequential objectives	Says nothing about value

Method	How it works	Best for	Weakness
Cost and latency	Do quick wins first; start slow items immediately	Every investigation, as a scheduling overlay	Confuses cheapness with importance if used alone
Risk	Rank by the consequence of not knowing	Safety, security, and harm-prevention contexts	Tends to inflate; everything looks risky
Legal or regulatory deadline	Externally imposed order	Notification obligations, statutory windows	Non-negotiable, so it sits above the others

In practice, a good priority list is built from three overlaid methods: decision impact as the primary ranking, perishability as an override, and cost and latency as a scheduling layer that starts slow items on day one regardless of rank.

1.4.3 The Perishability Override

This is the one that catches students out. Some evidence disappears whether or not you are ready for it.

Evidence type	Typical perishability	Consequence of delay
Marketplace and social listings	Days to weeks; sellers delete and platforms remove	The primary evidence of the conduct is gone
Web pages generally	Weeks to months; edits are silent	You are left with the current version, not the one that mattered
Platform account profiles	Days	Identity attribution evidence lost
System and network logs	Days to months, by retention policy	The window closes permanently
CCTV	Days to weeks, by system	Same
Volatile system state	Minutes	Gone on reboot
Court and corporate records	Years; effectively stable	Little urgency
Property and permit records	Stable	Little urgency
Published news and books	Stable	Little urgency

The rule: capture perishable evidence first, even when it ranks lower on decision impact. A high-impact records search that can be done in week four is not urgent. A listing that will be deleted this week is, regardless of where it sits in the ranking.

1.4.4 A Worked Priority Matrix

Task	Decision impact	Perishability	Latency	Composite priority	Rationale
Capture all in-	High	Very high	Low	1	Perishability

Task	Decision impact	Perishability	Latency	Composite priority	Rationale
scope listings					override; primary evidence
Capture seller profile pages	Medium	Very high	Low	2	Same
Preserve and hash captures	High	n/a	Low	3	Without this the captures are not evidence
Initiate records requests to the county	Medium	Low	High (3 weeks)	4	Latency override; start now or it arrives late
Registry and entity expansion searches	High	Low	Low	5	High impact, stable, do early but not first
Court docket searches	Medium	Low	Medium	6	
Document appearance history	High	Medium	Medium	7	Some appearances may be removed
Property records on identified addresses	Low	Low	Low	8	
Trade press and industry background	Low	Low	Low	9	Fill time while slow items are pending
Historical pricing analysis	Low	Low	Medium	Descoped if short	Explicitly named as the first thing dropped

The bottom row is the point. Naming in advance what gets dropped converts a resource shortfall from a crisis into a plan.

1.4.5 Reprioritization

Priorities change. The discipline is that changes are recorded:

Trigger	Action
A finding raises the impact of a lower-ranked task	Reprioritize, record the trigger and the new order
Evidence proves more perishable than assumed	Escalate immediately; record it
A slow item returns early or late	Adjust the schedule; record it
Resources are reduced	Apply the stated drop order; inform the consumer in writing
The consumer's decision changes	This is a plan amendment, not a reprioritization

1.5 Investigation Plan Evidence Collection

1.5.1 What the Evidence Collection Plan Adds

Chapter 4 covered collection planning as a process artifact: indicators, sources, queries, coverage boundaries. This chapter adds the four elements that make collected material into **evidence**:

Element	Question
Collection method	How, specifically, will this be captured, and does the method preserve what matters?
Documentation requirement	What must be recorded at the moment of collection?
Legal and authorization basis	What permits us to collect this, and what constrains how?
Preservation procedure	How will it be kept unaltered, and how will that be demonstrable later?

A collection plan without these produces material. With them, it produces evidence.

1.5.2 Evidence Types and Collection Methods

Evidence type	Appropriate collection method	Critical detail	Common error
Web page (listing, profile, article)	Full-page capture (image and, where possible, saved page or PDF), plus the URL, timestamp with time zone, and the capturing account state	Capture the whole page including URL bar and system clock where the tool allows; also archive independently	Cropped screenshot with no URL and no timestamp
Document (PDF, filing, report)	Download the original file; do not print to a	Preserves metadata (Chapter 3) and the	Re-saving or re-exporting, which

Evidence type	Appropriate collection method	Critical detail	Common error
	new PDF	original hash	destroys metadata and changes the hash
Image or video from a page	Download the original file where available, not a screenshot of it	Retains any surviving metadata and the original encoding	Screenshotting an image, which discards everything
Structured data (registry results, CSV export)	Export in the native format offered; also capture the query, filters, and result count	Reproducibility depends on recording the query	Pasting results into a document with no query record
Public record obtained in person or by request	Obtain a certified or official copy where available; record the request, the custodian, and the response	Certification matters if the record may be used formally	Photographing a screen at a terminal with no provenance record
Email or message provided by a client	Obtain the original file with headers, not a forwarded copy or a screenshot	Headers are the evidence; forwarding rewrites them	Accepting a printed copy
Log data	Obtain an export by an authorized administrator, with the export parameters recorded	Retention windows mean speed matters	Requesting after the retention period
Physical item	Photograph in place, record location and condition, package, seal, label, and log	Handling is the whole game	Any handling before documentation
Analyst-generated artifact (timeline, chart, notes)	Version, date, and attribute to the analyst	This is work product, not evidence, and must be labelled as such	Filing analytic products in the evidence inventory

The single most consequential line in that table is the second: download the original file rather than printing to a new one. A student who "saves" a filing by printing to PDF has produced a different file, with different metadata and a different hash, from the one that exists on the source system. It is still useful; it is no longer the original.

1.5.3 Physical and Digital Evidence

Both appear in OSINT-adjacent investigations, and their handling differs fundamentally.

Dimension	Physical evidence	Digital evidence
Uniqueness	The item is unique; there is one of it	Perfect copies are possible and normal
What is preserved	The object itself	The bit stream, verified by hash
Alteration	Usually visible: damage, wear, contamination	Usually invisible: a single byte changes nothing you can see
Integrity demonstration	Seals, packaging, storage, and the custody log	Hash values, plus the custody log
Working with it	Working with the original risks degradation	Work from a verified copy; the original is preserved untouched
Contamination risk	Physical transfer, environment, mishandling	Metadata changes from opening, indexing, antivirus, cloud sync
Volatility	Low, once secured	Can be high; remote content changes or disappears
Storage	Secure physical facility, controlled access, environmental controls	Write-protected storage, access control, backups, integrity checks
Transfer	Physical handover with signature	File transfer plus hash verification on receipt
Duplication	Photography or forensic replication; never identical	Bit-for-bit identical copies are routine
Typical challenge	"How do we know this is the item you seized?"	"How do we know this file is unaltered since collection?"
The answer	An unbroken custody chain and an intact seal	An unbroken custody chain and a matching hash

The practical implication for OSINT work: almost everything you handle is digital, which means the hash is your seal and the custody log is your packaging. Neither is optional.

1.5.4 The Evidence Collection Plan Row

Field	Content
Plan row ID	
Objective ID	Why this is being collected
Evidence sought	What, specifically
Source and system	Chapter 3 category plus the specific system
Collection method	From the table in 1.5.2
Perishability	And therefore urgency
Legal and authorization basis	Four tests result, plus any client authorization

Field	Content
	required
Documentation required at collection	The specific fields
Preservation procedure	Hash, storage location, backup, verification schedule
Who collects	
Expected coverage boundary	Written in advance
Status	

1.5.5 Preservation

Preservation is keeping evidence in the state it was collected, and being able to demonstrate that later.

The core practices for digital material:

1. **Hash at acquisition.** Compute a cryptographic hash (SHA-256 is the current standard for this work) immediately on collection, and record it in the evidence log. The hash is the fingerprint.
2. **Store a preservation copy that is never opened.** Working copies are made from it. The preservation copy is not browsed, edited, indexed, or opened.
3. **Work from a copy.** Every analytic action happens on a working copy, which may be freely marked up.
4. **Verify on transfer and on schedule.** Re-hash on every transfer and at defined intervals; record each verification, including successful ones.
5. **Control access.** Who can reach the store, and is access logged?
6. **Back up, and verify the backup.** A backup that has never been restored is a hope.
7. **Record retention and disposal.** What is kept, for how long, and how it is destroyed.

What a hash does and does not establish. A matching hash establishes that the file is bit-for-bit identical to the file that was hashed at acquisition. It does not establish that the content is truthful, that the source was authentic, that the collection was authorized, or that the file was not already altered before you collected it. It answers exactly one question, and it answers it conclusively.

1.5.6 Legal and Ethical Elements in the Evidence Plan

Carried forward from Chapters 2 through 4 and made concrete here:

- The four tests (access, identity, protection, purpose and harm) applied per row, before collection.
- The prohibited-activity statement.
- The individuals policy.

- Authorization: what requires client, counsel, or supervisory approval before it is done. In the labs, making a purchase and initiating contact both require written authorization, and both are excluded by default.
 - Terms of service positions for each platform in scope.
 - Handling of anything encountered that is out of scope or not lawfully public.
 - Retention and disposal.
-

1.6 Investigation Plan Documentation

1.6.1 Why Documentation Is Substantive

Rule 7 states that if it is not written down, it did not happen. Three reasons this is a claim about the quality of the work rather than about paperwork:

Reproducibility. An investigation that cannot be reproduced cannot be verified, and an unverifiable finding is an opinion. Documentation is what makes reproduction possible.

Memory is unreliable and gets worse. Six weeks after a search, you will not remember which term set you used, which filters were applied, or how many results were returned. You will believe you remember, which is worse.

Challenge is normal. Every investigation that matters is eventually questioned. The questions target process, and only documentation answers them.

1.6.2 The Documentation Set

Artifact	Purpose	Created when
Investigation plan	Authority, scope, objectives, priorities, procedures	Before work; amended by version
Investigator notes	Contemporaneous record of what was done, when, and why	Continuously, at the time
Evidence log / inventory	Every item, with source, hash, and custody reference	At each acquisition
Chain of custody records	Every transfer and verification	At each event
Search log	Every query: system, terms, filters, date range, result count, timestamp	At each search
Capture records	For each screenshot or page capture: URL, timestamp, tool, capturing conditions	At each capture
Decision log	Significant investigative decisions, alternatives, reasons	At each decision

Artifact	Purpose	Created when
Lead register	Every lead, its source, status, and disposition	Continuously
Timeline	From Chapter 5	Continuously
Analytic work product	Hypothesis matrices, link charts, assumption registers	During analysis
Communications record	Requests, approvals, client instructions	At each exchange
Report and annexes	The product	At the end
Closure record	What was accomplished, what remains, retention and disposal	At closure

Note the distinction between evidence and work product. Evidence is material collected from the world. Work product is what the investigator produced. They live in separate parts of the case file, and mixing them is a recurring error that makes an evidence inventory unreviewable.

1.6.3 Investigator Notes

The most basic and most neglected artifact. A professional standard:

Property	Requirement
Contemporaneous	Written at the time, not reconstructed later. Late entries are marked as late, with the date of entry
Dated and timed	Every entry, with time zone
Attributed	Who wrote it
Sequential	Numbered or paginated; no gaps
Factual	What was done and observed, separate from what was concluded
Complete on negatives	Searches that found nothing are recorded with their coverage boundary
Corrections visible	Errors are struck through and initialled, not deleted or overwritten
Non-destructive format	Append-only where possible; if a document, versioned

On corrections. The instinct is to fix an error silently. The professional practice is the opposite: a visible correction with a date and initials is evidence of a careful record, while a silently altered note undermines the whole file if it is ever noticed.

A worked note entry:

Entry 47. 12 May, 14:22 CDT. RM. Searched [platform B] using term set TS-2 (recorded in search log SL-018), filters: category "outdoor equipment," ships to US, active listings only. Result count 34. Compared against inventory; 9 listings not previously captured. Captured all 9 (items E-041 through E-049), hashed and logged. Note: 3 listings previously captured (E-012, E-017, E-023) no longer appear in results; preservation copies retained; cause of removal unknown and not determinable from public information. **Observation, not conclusion:** removal of previously captured listings is consistent with seller deletion, platform action, or a search index change. No basis to distinguish at this time.

That entry records what was done, what was found, what was not found, what changed since last time, and explicitly separates observation from conclusion. It takes ninety seconds to write and it is the difference between a defensible file and a folder of images.

1.6.4 Screenshots and Captures

Captures are the most common evidence type in OSINT work and the most commonly done badly.

Requirement	Reason
Full page, not a crop	A crop invites the question of what was outside it
URL visible or recorded	Otherwise the item is unattributable
System date and time visible or recorded, with time zone	Establishes when
Capture tool and version recorded	Reproducibility
Scroll capture or multiple sequential captures for long pages	Completeness
Original file downloaded where the item is a file	Metadata preservation
Independent archive where possible	Corroborates the capture
Hashed immediately	Integrity
Logged with item ID	Traceability
No annotation on the original	Annotate a working copy only

That last line matters. An arrow drawn on the original capture creates a new file, changes the hash, and destroys the original. Annotate a copy, and label it as an annotated working copy.

1.6.5 Good and Poor Documentation

The difference is usually visible within thirty seconds of opening a case file.

Property	Good	Poor
Timing	Contemporaneous	Reconstructed at the end
Completeness	Negative results recorded	Only findings recorded
Attribution	Every entry signed and timed	Anonymous, undated

Property	Good	Poor
Reproducibility	Queries, filters, and tools recorded	"I searched online"
Separation	Facts distinct from conclusions	Narrative mixing both
Corrections	Struck through, dated, initialled	Silently overwritten
Naming	Consistent, item IDs throughout	"screenshot17final(2).png"
Structure	Standard templates	Free-form notes in varying formats
Storage	One case store with defined structure	Files across a desktop, email, and a phone
Integrity	Hashes recorded at acquisition	No hashes
Custody	Every transfer logged	No record of who had what
Review	Someone else has read it	Nobody but the author has ever opened it
Test	A stranger could reproduce the work	Only the author can explain it, and only recently

1.6.6 The Reproducibility Test

The single best documentation check, and it takes ten minutes:

Hand your case file to a colleague who has no context. Ask them to pick any three findings and, using only the file, determine where each came from, when it was collected, how, by whom, what was searched to find it, and whether it has been altered since.

If they cannot, the documentation has failed, and it has failed now rather than later when it matters. Run this test at the midpoint of every investigation, not at the end.

1.7 Investigation Plan Chain of Custody

1.7.1 What Chain of Custody Is

Chain of custody is the documented, unbroken record of every person who had control of an item of evidence, from acquisition to final disposition, with every transfer, storage location, and verification recorded.

Its purpose is to answer one question convincingly: **is this the same item, in the same state, as when it was collected?**

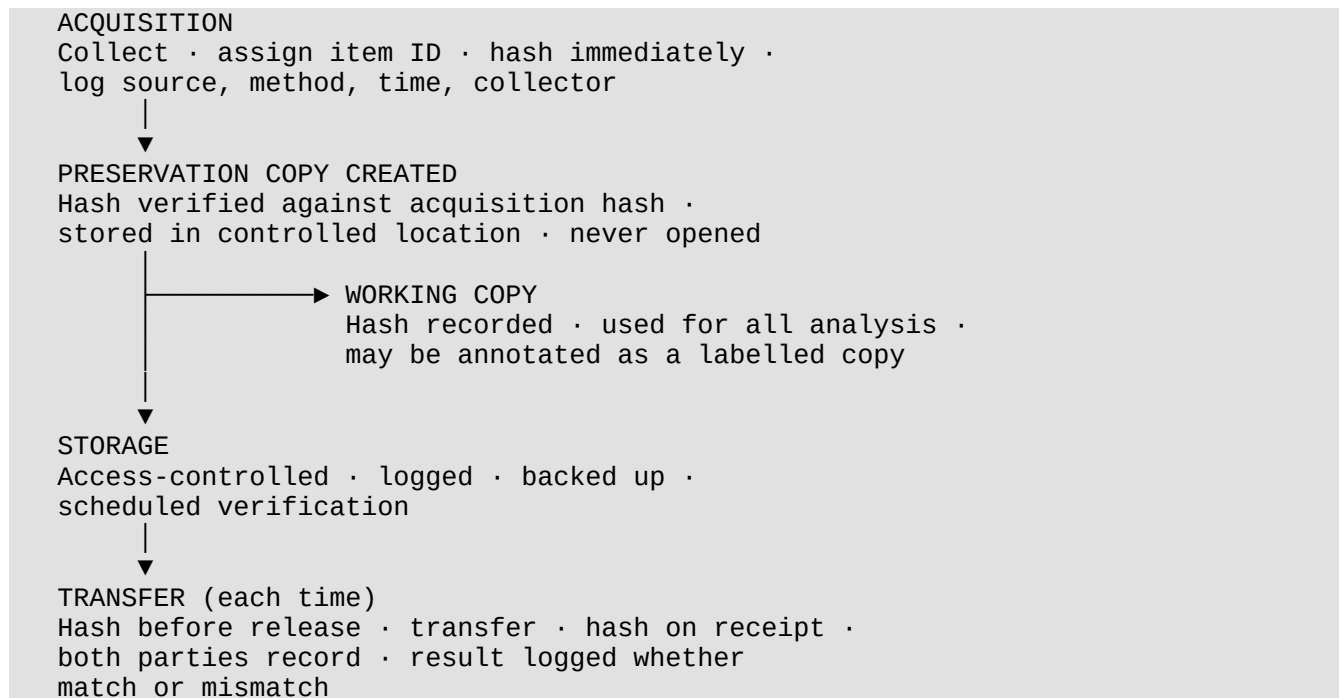
1.7.2 What a Custody Record Contains

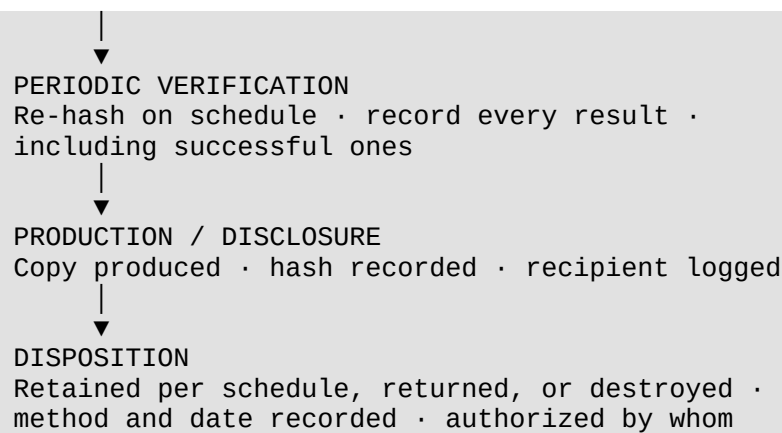
For each item, a continuous sequence of entries. Every entry:

Field	Content
Item ID	Unique, assigned at acquisition and never reused
Description	What it is, specifically
Event type	Acquisition, transfer, access, verification, duplication, storage change, disposal
Date and time	With time zone
From (person and role)	Who released it
To (person and role)	Who received it
Purpose	Why it moved
Method	How it moved: physical handover, encrypted transfer, copy
Hash before	For digital items
Hash after	For digital items
Verification result	Match or mismatch, recorded either way
Storage location after the event	Specific
Signatures or system attribution	Both parties
Notes	Condition, packaging, anomalies

The acquisition entry is the foundation. It records what was collected, from where, by whom, when, by what method, and the hash at the moment of acquisition. Everything after depends on it.

1.7.3 The Custody Lifecycle





1.7.4 What Breaks the Chain

Break	How it happens	Consequence
Gap in the record	An item is in one place on Tuesday and another on Friday with nothing in between	The item's state during the gap cannot be accounted for; the whole item becomes challengeable
Unlogged access	Someone opens the preservation copy	Metadata changes; the hash may or may not change; the record no longer matches reality
Hash mismatch	Verification fails	Either the item changed or the record is wrong; both are serious and must be reported
No acquisition hash	Hashing was done later	You can prove the item has not changed since you hashed it, and nothing about the period before
Shared credentials	Two people using one account	Attribution is impossible; the record cannot say who did what
Untracked copies	Copies made for convenience	Copies proliferate and no one can say which is the evidence
Working on the original	Analysis performed on the preservation copy	The original is altered; there is nothing to fall back to
Retroactive entries	The log written at the end	The log records the author's recollection, not the events
Anonymous entries	No attribution	The chain has a link nobody owns

1.7.5 When a Break Occurs

Breaks happen. What distinguishes a professional response is that a break is **reported, not repaired**.

The procedure:

1. **Stop using the item** for any purpose pending review.
2. **Record the discovery:** what was found, when, by whom, and how.
3. **Do not attempt to reconstruct the missing entries.** Reconstructed custody entries are worse than an acknowledged gap, because they misrepresent the record. If a gap exists, the record says a gap exists.
4. **Determine the extent:** which items, which period, which findings depend on them.
5. **Notify** the lead, the client, and counsel as the plan specifies.
6. **Assess the impact** on each affected finding, and state it in the product.
7. **Report the break in the final report.** An acknowledged break with a stated impact is survivable. A concealed break, discovered later, destroys the credibility of the entire investigation and the investigator.

A hash mismatch specifically. Record the expected hash, the computed hash, the file, and the time. Do not overwrite the log entry. Determine whether the mismatch reflects a changed file, a mis-transcribed hash, a different algorithm, or a re-saved copy. Report the finding either way. In Lab 4 you will encounter exactly this and the grading rewards correct reporting rather than a clean-looking log.

1.7.6 Proper and Improper Custody in Practice

Practice	Proper	Improper
Item identification	Unique ID assigned at acquisition, never reused	Files named by content, IDs reused across cases
Hashing	At acquisition, before anything else	When someone remembers, or at the end
Preservation copy	Created, hashed, never opened	The only copy is the one being worked on
Access	Controlled and logged	Shared folder, open access
Transfers	Hashed both ends, both parties record	Emailed with no record
Annotation	On labelled working copies only	On the original
Log entries	Contemporaneous, attributed, corrections struck through	Written at the end, unattributed, edited
Verification	Scheduled, results recorded whether pass or fail	Only when something looks wrong
Breaks	Reported with impact assessed	Quietly reconstructed
Disposal	Authorized, dated, method recorded	Deleted when the project folder was cleaned up

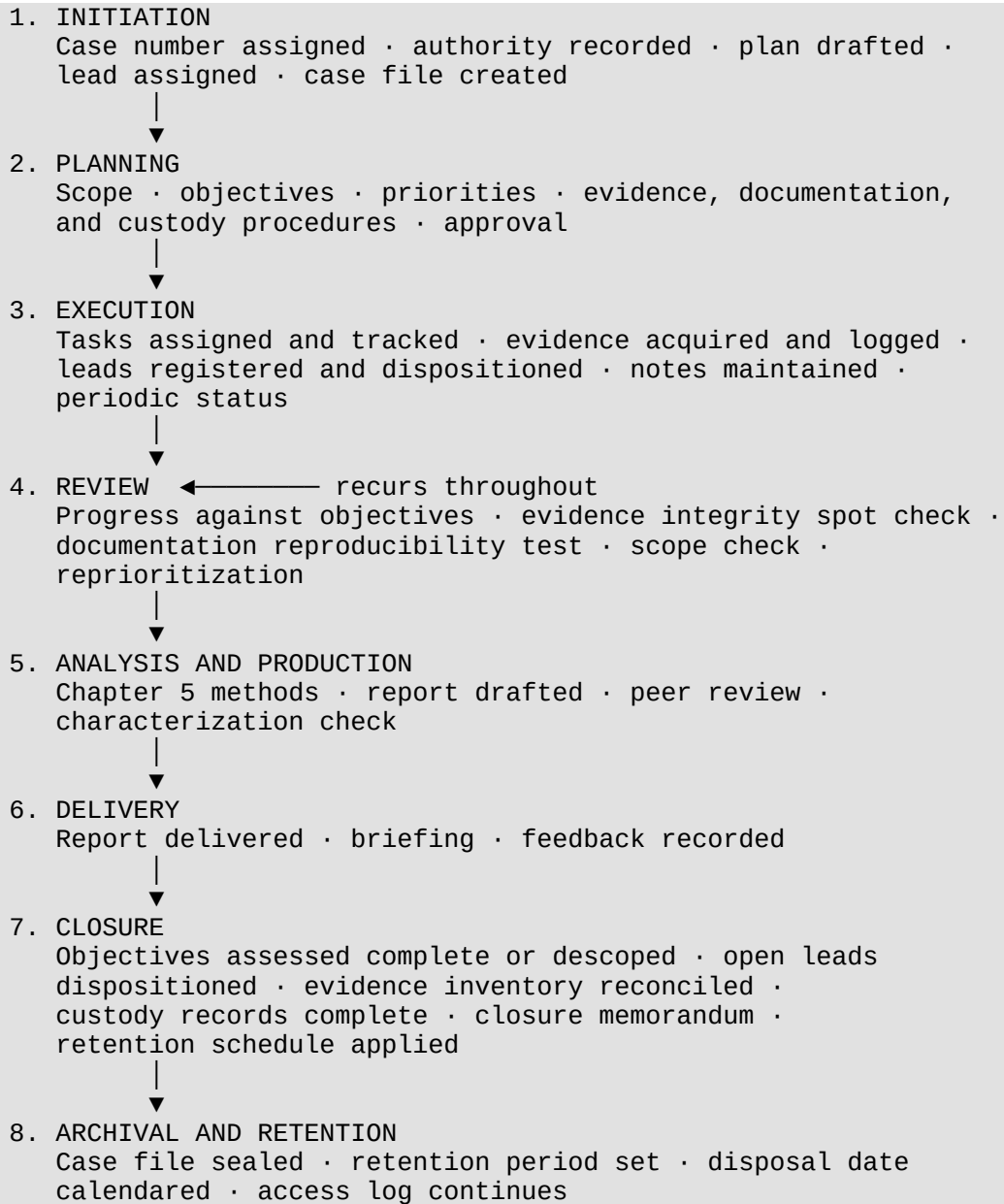
1.8 Investigation Plan Case Management

1.8.1 What Case Management Is

Case management is the discipline of running the investigation as an organized effort: assigning and tracking work, managing leads, controlling the case file, coordinating people, reviewing progress, and bringing the case to a defined close.

Planning says what will be done. Case management is how it actually gets done, tracked, and finished.

1.8.2 The Case Lifecycle



1.8.3 The Case File

One place, with a defined structure, containing everything. A workable structure for an individual or

small-team investigation:

```
CASE-2026-0142/
├── 00-administration/
│   ├── authorization.pdf · plan-v1.docx · plan-v2.docx ·
│   ├── approvals.pdf · communications-log.xlsx
├── 01-planning/
│   ├── scope-statement · objectives-register ·
│   ├── priority-matrix · evidence-collection-plan
├── 02-evidence/
│   ├── inventory.xlsx
│   ├── └── preservation/      (never opened; one folder per item)
│   │   └── working/          (copies for analysis)
├── 03-custody/
│   ├── custody-log.xlsx · transfer-forms/ · verification-records/
├── 04-documentation/
│   ├── investigator-notes.docx · search-log.xlsx ·
│   ├── capture-records.xlsx · decision-log.xlsx ·
│   └── lead-register.xlsx
├── 05-analysis/
│   ├── timeline.xlsx · link-chart · edge-register ·
│   └── hypothesis-matrix · assumption-register
├── 06-products/
│   ├── draft-report · final-report · annexes · briefing
├── 07-closure/
│   ├── closure-memorandum · retention-record ·
│   └── open-leads-disposition
```

Two features matter more than the specific structure. Preservation and working copies are separated at the folder level, so opening the wrong one requires a deliberate act. And administration, evidence, documentation, analysis, and products are separated, so an evidence inventory is not polluted with work product.

1.8.4 Task and Lead Management

A task is work to be done, derived from an objective. **A lead** is a possibility to be pursued, generated during the work. They are tracked differently.

	Task register	Lead register
Origin	Derived from an objective in the plan	Generated by findings during execution
Fields	Task ID, objective ID, description, owner, due, status, output	Lead ID, source item, description, potential value, in scope?, priority, status, disposition
Status values	Not started, in progress, blocked, complete, descope	Open, being worked, resolved, closed as unproductive, out of scope, referred
Closure requirement	The output exists	A recorded disposition, always
The failure	Tasks with no objective	Leads that quietly disappear

Every lead gets a disposition. "Closed as unproductive, no further action; basis: searched X and Y, nothing found, coverage boundary recorded" is a disposition. A lead that stops being mentioned is not closed, it is lost, and at closure someone will ask what happened to it.

1.8.5 Small and Large Investigations

Dimension	Small (one investigator, days)	Large (team, months)
Plan	One page	Fifteen to forty pages with annexes
Roles	One person does everything	Lead, collectors, analyst, evidence custodian, reviewer, counsel liaison
Case file	A folder with the structure above	A managed repository with access control
Task tracking	A list	A tracked board with owners and dependencies
Evidence custody	Self-logged	A designated custodian; investigators do not hold evidence directly
Review	Self-review after a break, named as a weaker control	Scheduled peer and supervisory review
Communication	Ad hoc with the consumer	Defined cadence, single point of contact, written status
Deconfliction	Not needed	Required: who is working what, to avoid duplicate contact and duplicate collection
Closure	A memo	Formal closure with reconciliation and sign-off
Biggest risk	Nothing is written down	Nobody knows what anyone else has

The scaling insight: small investigations fail from missing documentation; large ones fail from missing coordination. The plan sections that matter most differ accordingly.

1.8.6 Manual and Software-Based Case Management

Dimension	Manual (documents and spreadsheets)	Software-based (case management platform)
Cost	Effectively zero	Licensing, administration, training
Setup time	Minutes	Weeks, including configuration
Discipline required	Very high; the system does not	Lower; the tool enforces fields

Dimension	Manual (documents and spreadsheets)	Software-based (case management platform)
	enforce anything	and workflow
Audit trail	Only what you record	Automatic, including who viewed what and when
Access control	Filesystem permissions at best	Role-based, granular, logged
Concurrent users	Poor; conflicting versions	Designed for it
Search and retrieval	Manual	Indexed
Integrity support	Manual hashing and logging	Frequently built in, with automatic verification
Reporting	Manual assembly	Generated
Portability	High; the files are the files	Low; export formats may be limited
Failure mode	Discipline lapses and nothing catches it	Complacency: the tool is assumed to be doing something it is not configured to do
Appropriate for	Individual work, small teams, short cases, learning	Teams, ongoing caseloads, regulated environments, anything that will be litigated

Both work. A disciplined investigator with spreadsheets produces a better case file than an undisciplined one with a platform. The labs in this chapter use the manual approach deliberately, because doing it by hand teaches what the software is doing, and because a practitioner who only knows the tool cannot tell when the tool is misconfigured.

1.8.7 Closure

Closure is a defined act, not the moment the report is sent. A case is closed when:

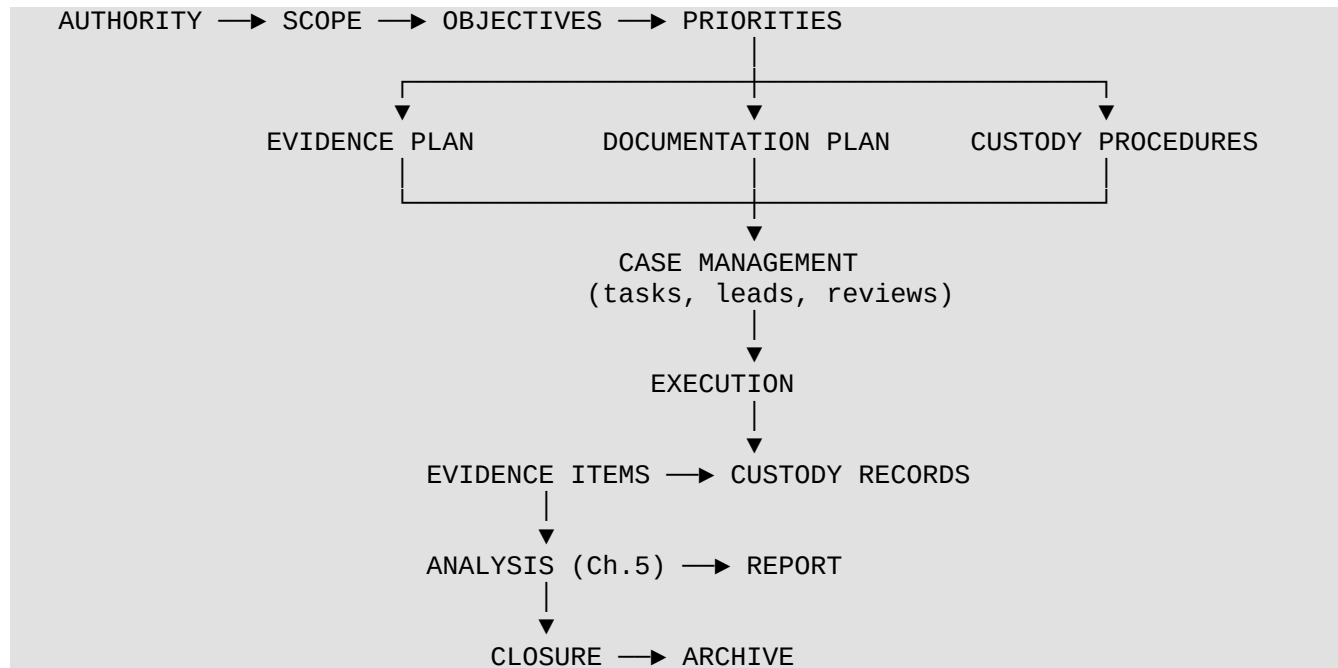
1. Every objective is marked complete, descoped, or blocked, with a stated basis.
2. Every lead has a recorded disposition.
3. The evidence inventory reconciles: every item logged is present, every item present is logged.
4. Custody records are complete and every item's last entry states its current location.
5. Documentation passes the reproducibility test.
6. The report is delivered and receipt is confirmed.
7. Feedback is recorded (Chapter 4).
8. A closure memorandum records what was accomplished, what was not, what remains open, and any custody or documentation exceptions.

9. Retention is applied: what is kept, for how long, and the disposal date is calendared.

The closure memorandum is the document a future reader will find first. It should be honest about what was not accomplished. An investigation that met four of six objectives and says so is more useful, and more credible, than one that quietly reports only what it achieved.

1.9 Putting It Together

1.9.1 The Planning Artifact Chain



The traceability test for this chapter: pick any evidence item in the file. You should be able to state which plan row authorized its collection, which objective that row served, which scope inclusion permitted it, who collected it, when, how, its acquisition hash, every subsequent custody event, and which finding it supports. If any link is missing, that item is not defensible, and any finding resting on it inherits the defect.

1.9.2 The Plan Review Checklist

Before a plan is approved, someone other than its author should confirm:

- The authority is recorded and adequate.
- All four scope dimensions are bounded.
- Exclusions exist and are specific.
- Every objective is measurable and completable by a negative finding.
- Every objective has a success criterion and a completion test.

- Priorities are ranked by a stated method, with a drop order.
 - Perishable evidence is prioritized ahead of stable evidence.
 - Slow items are scheduled first.
 - Every evidence plan row has a collection method, a legal basis, a documentation requirement, and a preservation procedure.
 - The documentation plan names the artifacts, formats, and storage.
 - Custody procedures cover acquisition, preservation, working copies, transfer, verification, and disposal.
 - Roles are assigned, including who holds evidence.
 - Review points are scheduled.
 - Closure criteria are stated.
 - The scope change procedure is written and its default is stop.
-

Part 2: Comparison Tables

2.1 Scope vs. Objectives vs. Priorities

Dimension	Scope	Objectives	Priorities
Answers	Where may we look?	What must we accomplish?	In what order, and what gets dropped?
Form	Inclusions, exclusions, boundaries, assumptions	Measurable statements with success criteria	A ranked list with a method and a drop order
Bounded by	Subject, time, geography, activity	The strategic goal and the scope	Resources and the deadline
Number	One statement	Three to seven	One ranking covering all tasks
Changes how	Only by approved amendment	Only by approved amendment	Reprioritization, recorded but not requiring amendment
Owner	The consumer, with the lead	The consumer, with the lead	The lead
Constrains	What objectives are possible	What tasks are needed	What order tasks run in
Failure	Unbounded, or no exclusions	Unmeasurable, or completable only by a positive finding	Unranked, or no drop order

Dimension	Scope	Objectives	Priorities
Consequence of failure	Scope creep; legal exposure; unlimited effort	Work without completion	Effort spent on the wrong things by the deadline
Reviewer's test	Can I tell what is out?	Can two people agree it is done?	Can I see what happens if resources fall short?
Artifact	Scope statement	Objectives register	Priority matrix

2.2 Strategic Goals vs. Tactical Tasks

Dimension	Strategic goal	Objective	Tactical task
Number	One	Three to seven	Dozens
Statement	Why the investigation exists, tied to a decision	What must be accomplished	What work will be performed
Duration	The whole investigation	Days to weeks	Hours to days
Owner	Consumer with the lead	Named objective owner	Individual investigator
Changes	Rarely, by amendment	By amendment	Constantly
Measured by	Whether the decision was supported	Whether the success criterion is met	Whether the output exists
Example	Give counsel the evidence base to decide by 30 June among enforcement, civil action, or referral	Produce a captured, hashed, logged inventory of all in-scope listings	Search platform B with term set TS-2 and capture new listings
Written	In the plan's opening section	In the objectives register	In the task register
Failure signature	Investigation succeeds; the decision is unsupported	Objective complete; nobody can say what it contributed	Task complete; it served no objective
Aggregation	Achieved through objectives	Achieved through tasks	Produces an output

2.3 Evidence Types and Appropriate Collection Methods

Evidence type	Appropriate method	Must be recorded	Preservation	Common error
Web page or listing	Full-page capture plus saved page or PDF, plus independent	URL, timestamp with zone, tool and version, account state, capture	Hash at capture; preservation copy untouched	Cropped screenshot, no URL, no timestamp

Evidence type	Appropriate method	Must be recorded	Preservation	Common error
	archive	conditions		
Downloadable document	Download the original file	Source URL, download time, file name, size, hash	Hash at acquisition; do not re-save	Printing to a new PDF, destroying metadata and changing the hash
Image or video on a page	Download the original media file	Source URL, time, hash	As above	Screenshotting the image
Structured query results	Native export plus the query record	System, exact query, filters, date range, result count, timestamp	Hash the export	Pasting results with no query record
Public record (portal)	Download the official copy; request certified copy if it may be used formally	Custodian, record identifier, retrieval time	Hash; retain the request record	Photographing a terminal screen
Public record (in person or by request)	Official or certified copy	Request date, custodian, response date, fees, exemptions applied	Hash any digital copy; retain the paper original	No record of the request itself
Client-provided email	Original file with full headers	Provider, custodian, transfer method, hash	Preservation copy; never edited	Accepting a forward or a printout
Log export	Export by an authorized administrator	Export parameters, system, time range, operator, retention policy	Hash on receipt	Requesting after retention expiry
Physical item	Photograph in place, then package, seal, label, log	Location, condition, seal number, collector	Secure storage, environmental control	Handling before documenting
Analyst work product	Version, date, attribute	Author, version, date, inputs used	Version control	Filing it in the evidence inventory

2.4 Good Documentation vs. Poor Documentation

Property	Good	Poor	Test
Timing	Written at the time	Reconstructed at the end	Do timestamps cluster on one day?
Attribution	Every entry signed and	Anonymous or undated	Can you tell who did

Property	Good	Poor	Test
	timed		this?
Negatives	Recorded with coverage boundaries	Only findings recorded	Are there any "found nothing" entries?
Reproducibility	Queries, filters, tools, and versions recorded	"Searched online"	Could a stranger repeat it?
Separation	Observation distinct from conclusion	Narrative mixing both	Can you find the facts without the interpretation?
Corrections	Struck through, dated, initialled	Silently overwritten	Is there any evidence of correction at all?
Naming	Item IDs used consistently	Descriptive or duplicate file names	Does the same item have one name everywhere?
Structure	Standard templates, consistent fields	Free-form, varying	Do two entries look alike?
Storage	One case store with a defined structure	Scattered across devices and email	Is there one place?
Integrity	Hashes at acquisition	None	Can you prove nothing changed?
Custody	Every transfer logged	No transfers recorded	Who has had this?
Completeness	Plan, notes, evidence log, search log, decision log, lead register all present	Screenshots and a report	Which artifacts are missing?
Review	Read by someone other than the author	Never opened by anyone else	Has a second person read it?

2.5 Physical Evidence vs. Digital Evidence

Dimension	Physical	Digital
Uniqueness	One of it exists	Perfect copies are normal
Integrity mechanism	Seal, packaging, secure storage, custody log	Cryptographic hash plus custody log
Alteration visibility	Usually visible	Usually invisible
Working practice	Minimize handling of the original	Never work on the original; use a verified copy
Contamination sources	Transfer, environment, mishandling	Opening, indexing, antivirus, cloud sync, re-saving

Dimension	Physical	Digital
Volatility	Low once secured	Can be high; remote content changes or vanishes
Duplication	Never identical	Bit-for-bit identical
Storage requirements	Secure facility, access control, environment	Write-protected storage, access control, backup, scheduled verification
Transfer	Physical handover with signatures	File transfer plus hash verification both ends
Typical challenge	"Is this the item you collected?"	"Is this file unaltered since collection?"
The answer	Intact seal plus unbroken custody chain	Matching hash plus unbroken custody chain
Recovery from a break	Sometimes impossible	Sometimes possible if a verified preservation copy exists
Relevance to OSINT work	Occasional	Nearly everything

2.6 Proper vs. Improper Chain of Custody

Practice	Proper	Improper	Why it matters
Item ID	Unique, assigned at acquisition, never reused	Reused, or files named by content	Ambiguous identity defeats the whole record
Hashing	At acquisition, before any other action	Later, or never	A late hash proves nothing about the earlier period
Preservation copy	Created, verified, never opened	Only one copy, which is worked on	No fallback if the working copy is altered
Working copies	Made from the preservation copy, hashed, labelled	Undifferentiated copies	Nobody can say which is the evidence
Access	Controlled and logged	Open shared folder	Unlogged access is an unaccountable gap
Attribution	Individual accounts and signatures	Shared credentials	The chain has a link nobody owns
Transfers	Hashed both ends, both parties record	Emailed with no record	The transfer period is unaccounted for
Annotation	On labelled working copies only	On the original	Alters the original and changes the hash
Log entries	Contemporaneous,	Retroactive,	The log records

Practice	Proper	Improper	Why it matters
	attributed, corrections visible	unattributed, edited	recollection, not events
Verification	Scheduled, all results recorded	Only when something looks wrong	Successful verifications are the evidence of continuity
Breaks	Reported, extent determined, impact stated	Reconstructed or concealed	A concealed break destroys the whole investigation's credibility
Disposal	Authorized, dated, method recorded	Deleted during cleanup	Unauthorized destruction is its own problem

2.7 Small vs. Large Investigation Plans

Element	Small (one investigator, days)	Large (team, months)
Plan length	One page	Fifteen to forty pages plus annexes
Authority	One line	Formal tasking with cited authority
Scope	Three to five bullets covering four dimensions	Full statement with change control procedure
Objectives	Two to three	Five to seven with an objective hierarchy
Priorities	Numbered list with a drop order	Weighted matrix with contingency and resource allocation
Evidence plan	Six to twelve rows	Dozens of rows with per-row legal review
Documentation	Notes, evidence log, search log	Full standard with templates, storage architecture, retention schedule
Custody	Self-logged with hashes	Designated custodian; investigators do not hold evidence
Roles	One	Lead, collectors, analyst, custodian, reviewer, counsel liaison
Review	Self-review after a break, named as a weaker control	Scheduled peer and supervisory review

Element	Small (one investigator, days)	Large (team, months)
Communication	Ad hoc	Defined cadence, single point of contact, written status
Deconfliction	Not required	Required
Closure	A memorandum	Formal reconciliation and sign-off
Dominant risk	Nothing is written down	Nobody knows what anyone else has
Where the plan effort goes	Scope and documentation	Coordination, custody, and review

2.8 Manual vs. Software-Based Case Management

Dimension	Manual	Software-based
Cost	Effectively zero	Licence, administration, training
Time to start	Minutes	Weeks including configuration
Discipline required	Very high; nothing is enforced	Lower; fields and workflow are enforced
Audit trail	Only what is recorded manually	Automatic, including views and edits
Access control	Filesystem permissions	Role-based, granular, logged
Concurrency	Poor; version conflicts	Designed for teams
Retrieval	Manual	Indexed search
Integrity support	Manual hashing and logging	Often built in with scheduled verification
Reporting	Manually assembled	Generated
Portability	High	Limited by export capability
Learning value	High; you see what the process requires	Low; the process is hidden
Failure mode	A discipline lapse goes uncaught	Complacency: assuming the tool does something it is not configured to do
Best for	Individuals, small teams, short cases, training	Ongoing caseloads, teams, regulated or litigated work

2.9 Common Investigation Planning Mistakes and Their Consequences

Mistake	Where it originates	Immediate effect	Consequence	Prevention
No written plan	Initiation	Work starts fast	Nothing to review against; no defence to process challenge	A one-page plan, always
No recorded authority	Initiation	Nobody notices	The first question under challenge is unanswerable	Record who authorized what, when
Scope with no exclusions	Scope	Feels complete	Every adjacent thread appears permitted; drift is invisible	Write exclusions and the reason for each
Only two scope dimensions bounded	Scope	Looks like a scope	Expansion through the unbounded edges	Bound subject, time, geography, activity
No scope change procedure	Scope	Nothing until a thread appears	Out-of-scope collection with no authorization	Default-stop procedure, written
Unmeasurable objectives	Objectives	Sounds ambitious	Cannot report completion; the investigation never ends	Success criterion plus completion test per objective
Objectives completable only by a positive finding	Objectives	Seems normal	The investigator is implicitly paid to find something	Phrase as "determine whether," not "establish that"
Task list with no objectives	Objectives	Everyone is busy	Work complete, accomplishment unknown	Every task cites an objective ID
Unranked priorities	Priorities	Everything is important	At the deadline, the wrong work is done	Ranked list with a stated method
No drop order	Priorities	Nothing until resources shrink	The deadline decides for you	Name what is dropped first
Perishable evidence not prioritized	Priorities	Orderly progress	The primary evidence is deleted before capture	Perishability override
Slow items started late	Priorities	Feels efficient	Discriminating evidence arrives after the decision	Initiate slow items on day one

Mistake	Where it originates	Immediate effect	Consequence	Prevention
No collection method specified	Evidence plan	Material is gathered	Screenshots without URLs; PDFs printed rather than downloaded	Method per row
No legal basis per row	Evidence plan	Collection proceeds	Unauthorized or terms-breaching collection; findings unusable	Four tests per row before collection
No hashing at acquisition	Custody	Nothing visible	Integrity cannot be demonstrated for the earlier period	Hash first, then everything else
Working on the preservation copy	Custody	Convenient	The original is altered; no fallback exists	Folder-level separation
Retroactive logging	Documentation	Saves time now	The log records recollection; the file is unauditible	Log at the time; mark any late entry
Negatives not recorded	Documentation	The file looks tidy	A limitation is later read as a conclusion	Record every negative with its coverage boundary
Annotating originals	Documentation	Clearer for the reader	Original altered, hash changed	Annotate labelled working copies
Evidence and work product mixed	Case management	Everything is in one place	The evidence inventory becomes unreviewable	Separate at the folder level
Leads not dispositioned	Case management	They fade quietly	At closure nobody can say what happened to them	Every lead gets a recorded disposition
No review points	Case management	Uninterrupted work	Errors compound to the end	Scheduled reviews including a reproducibility test
No closure criteria	Case management	The case drifts	It is never closed, and retention never starts	Closure checklist in the plan
Concealing a custody break	Custody	The file looks clean	Discovery later destroys the credibility of everything	Report, assess impact, state it in the report

Part 3: Case Studies

Each case walks the planning process, priorities, evidence strategy, documentation, custody, case management, challenges, and outcome. Cases marked *historical* describe real documented events in general terms; cases marked *composite* are built from documented professional practice and are not real organizations or people.

3.1 Cybersecurity Incident Investigation: A Payment Diversion at a Regional Nonprofit

(*composite*)

Situation. A regional nonprofit discovers that \$184,000 in vendor payments was diverted to an account it does not recognize. The finance director noticed a mismatched remittance advice. The executive director must decide within ten days whether to notify the board, the funder, the bank, and law enforcement, and what to tell each.

Planning. Authority recorded: the executive director engaged an external investigator with written authorization to examine the organization's own systems and public sources. Scope bounded to: the payment chain for three vendors over six months; the email accounts of the four staff in that chain; publicly available information on the receiving account's named holder; and the organization's own network and mail logs. Excluded: staff personal devices and accounts, any staff member's conduct outside the payment chain, and any contact with the receiving account holder.

Objectives. Four, all measurable: (1) determine whether the organization's mail systems were accessed without authorization, with the log evidence supporting either finding; (2) reconstruct the payment change sequence with a sourced timeline; (3) determine what public records establish about the receiving account's named holder; (4) produce a notification-readiness assessment stating what is established, what is not, and with what confidence, by day eight.

Priorities. Log retention drove everything. Mail and network logs had a 30-day retention window and the earliest suspected activity was 26 days old, so log preservation was priority one and was executed within four hours of engagement. Bank record requests, with a five-day latency, went out on day one. Public records research on the receiving account holder, stable and non-perishable, ran last.

Evidence strategy. Logs exported by the organization's IT provider with the export parameters recorded, hashed on receipt. Original message files with full headers obtained rather than forwards. The forged remittance advice obtained as the original attachment, not a printout. Public records downloaded as official copies. Every item hashed at acquisition, preservation copies stored untouched.

Documentation. Contemporaneous notes from the first hour. A search log recording every query against the mail system. A decision log capturing the choice not to examine staff personal accounts and the reason. Capture records for every public record retrieved.

Custody. A single custodian (the external investigator), individual account attribution throughout, hashes at acquisition, verification on each transfer to counsel. One transfer to the bank's fraud unit was

logged with hashes both ends.

Case management. Ten-day timeline with a day-five review. Task register tied to the four objectives. Lead register with eleven leads, of which six were dispositioned as out of scope and recorded.

Challenges. Two. First, the log retention window meant that the earliest activity of interest was four days from expiry; had planning taken a day longer the evidence would have been gone. Second, the finance director had forwarded the suspicious remittance advice to three colleagues before the investigation began, creating four copies with different headers and no record of the original. The investigation obtained the original from the mail store, hashed it, and recorded the forwarding as a documented pre-investigation handling event rather than pretending it had not occurred.

Outcome. Unauthorized mail access was established from the logs. The payment change sequence was reconstructed. Public records identified the receiving account's named holder as a recently formed entity with no other trace, which was reported as an identification with a coverage boundary rather than as a conclusion about who benefited. The organization notified its bank on day three (early, because the log finding was available early), the funder and board on day eight with the full assessment, and law enforcement with a package that included the custody records. The bank recovered a portion of the funds.

Planning lesson. The four-hour log preservation decision was worth more than every other action in the investigation, and it was possible only because perishability was assessed during planning rather than discovered during execution.

3.2 Criminal Investigation: A Retail Burglary Series

(composite)

Situation. A county investigator is assigned eleven commercial burglaries over four months, previously worked as separate cases by different deputies. The supervisor wants a determination of whether they are linked and, if so, an investigative plan for the series.

Planning. This case illustrates a specific planning problem: **consolidating existing cases into one.** Authority came from a written supervisory assignment consolidating the eleven case numbers under a series case number, with the original case files remaining intact and referenced. Scope: the eleven incidents, their physical and digital evidence, the surrounding geography, and public and internal records concerning any suspect developed. Excluded: unrelated offences by any individual identified, which would be referred rather than absorbed.

Objectives. (1) Determine whether the incidents share a method, timing, or geographic pattern sufficient to treat them as a series, with the analysis documented; (2) reconcile the existing evidence inventories across eleven case files into one series inventory with verified custody for every item; (3) produce an event-based analysis of each incident's internal sequence and a cross-incident comparison; (4) identify investigative steps not taken in the original cases.

Priorities. Objective 2 came first and consumed a third of the effort, which surprised the supervisor.

The reason: several items across the original cases had incomplete custody records, and a series case built on items whose custody cannot be established is a series case that fails in court. Perishability drove the rest: two businesses' CCTV systems had 21-day retention and both had already expired, which became a documented negative finding and a lesson recorded for the county's response procedures.

Evidence strategy. Physical items were re-inventoried against the original property records, each verified by seal integrity and log continuity. Two items had custody gaps and were flagged. Digital items (CCTV exports, point-of-sale logs, access records) were re-hashed and compared against acquisition hashes where recorded; three had no acquisition hash, which was recorded as an unrecoverable limitation.

Documentation. A series case file was built with a reconciliation table mapping every item to its original case, its original custody record, and its verification status. The event-based analysis used one timeline per incident plus a comparison matrix.

Custody. The reconciliation identified: nine items with complete chains; two items with gaps of several days between the original deputy's collection and the property room entry; three digital items with no acquisition hash. All five exceptions were documented, the affected findings were identified, and the prosecutor was advised in writing at the start rather than at disclosure.

Case management. One series file, eleven referenced source files, a task register organized by objective, and a lead register consolidating leads from all eleven original cases, several of which had never been dispositioned.

Challenges. The consolidation surfaced that three leads in the original cases had been recorded and never worked, and that two of them pointed at the same commercial address. This was reported as a process finding to the supervisor alongside the case findings.

Outcome. Eight of eleven incidents were assessed as a series on the basis of method and timing, with two indeterminate and one excluded. Two of the three previously unworked leads produced the investigative direction that had been missing. The two custody-gap items were not relied upon for any finding, and the prosecutor's early notice meant the exceptions were managed rather than discovered.

Planning lesson. **Custody reconciliation is a planning objective, not an administrative chore.** An investigator who assumes inherited evidence has a clean chain will discover otherwise at the worst possible moment.

3.3 Fraud Investigation: Grant Expenditure at a Municipal Programme

(composite)

Situation. A city's internal audit function receives an anonymous complaint alleging that a grant-funded youth programme has reported expenditures that did not occur. The city manager must decide within eight weeks whether to refer the matter, recover funds, or close the complaint.

Planning. Scope bounded carefully because the complaint was anonymous and unspecific:

expenditures reported under three grant cycles; the supporting documentation the programme submitted; publicly available records concerning the vendors named in that documentation. Explicitly excluded: programme staff's personal finances, participants, and the identity of the complainant, which the plan stated would not be investigated.

Objectives. (1) Determine, for each reported expenditure above a threshold, whether supporting documentation exists in the programme's submissions and whether it is internally consistent; (2) determine whether each named vendor exists as an identifiable business entity in public records; (3) reconcile reported expenditure totals against the grant's reporting periods; (4) produce an assessment with confidence levels and a recommendation on referral.

Priorities. Objective 2 was ranked highest on decision impact because a vendor that does not exist is a discriminating finding, and it was also the cheapest. Objective 3 was ranked second because the previous chapter's lesson applies: a period-mismatched reconciliation is worthless, so the reconciliation had to be built on aligned periods before any arithmetic meant anything.

Evidence strategy. All expenditure documentation obtained as originals from the grants office with a custody transfer record. Vendor research entirely from public records: registry filings, trade name filings, licensing, and property records. Every registry result downloaded as an official copy, hashed and logged. Notably, the plan required that a vendor's absence from a registry be recorded as a negative finding with the systems searched and name variants used, never as "the vendor does not exist."

Documentation. Templates for everything: a per-expenditure worksheet, a per-vendor worksheet, a search log, and a reconciliation table. Standardization mattered because the volume was high (214 expenditures) and consistency was what made review possible.

Custody. The grants office's original documentation was transferred to audit with a signed inventory. Digital copies were made, hashed, and the originals returned to a controlled store with the transfer recorded both ways.

Case management. Eight-week schedule with reviews at weeks three and six. The week-three review triggered a reprioritization when entity expansion revealed that three of the named vendors shared an officer, which raised the priority of the relationship analysis.

Challenges. Two vendors could not be located in any registry under any name variant. The temptation was to report them as fictitious. The plan's documentation requirement forced the correct formulation: no entity matching these names was located in the state registry, the county trade name index, or the licensing database, using the listed variants, as of the search date, and the coverage boundary was stated. A third vendor turned out to be a sole proprietor with no registration obligation, which is exactly the benign explanation the disciplined wording preserved room for.

Outcome. Of 214 expenditures, 191 were supported and consistent. Nine had documentation inconsistencies attributable to clerical error. Fourteen, totalling \$61,000, named two vendors that could not be located in public records and had no supporting documentation beyond an invoice. The assessment recommended referral limited to those fourteen, recovery discussions on nine, and no action on the remainder. The city referred, and the anonymous complainant's identity was never sought.

Planning lesson. The scope exclusion protecting the complainant, and the documentation requirement forcing coverage boundaries on negative findings, both did substantive work. Neither was added for legal cover; both improved the accuracy of the result.

3.4 Investigative Journalism: A Newsroom Records Project

(historical pattern, described in general terms)

Situation. Newsrooms running document-heavy investigations operate planning and documentation practices that closely parallel formal investigative work, for the same reason: the work will be challenged, and by parties with resources.

Planning element	How it runs in a newsroom project
Authority	Editorial commissioning, with a budget and a publication window; legal review engaged at the outset for high-exposure projects
Scope	Narrowed deliberately, often more than reporters want, because a bounded question can be verified and a thematic one cannot
Objectives	Framed as verifiable propositions: not "expose the programme" but "determine how many contracts bypassed competitive bidding and under which exemptions"
Priorities	Records requests first, because latency is measured in months and denials require appeals; interviews later, once the documentary base exists
Evidence strategy	Original documents, not summaries; every document catalogued with its provenance; requests and responses retained including denials and exemptions applied
Documentation	A document register with provenance for every item; a fact-checking matrix in which every published assertion is mapped to its supporting documents
Custody	Source protection drives an unusual custody model: sensitive material is compartmented, access is minimized, and the record of who holds what is itself protected
Case management	A publication schedule, a legal review gate, a right-of-reply process treated as a methodological step, and a corrections pathway
Challenges	Records denials and appeals; source protection constraining what may be documented;

Planning element	How it runs in a newsroom project
	publication deadlines competing with verification
Outcome	Findings published with the documentary basis shown, and the fact-check matrix retained so that any challenged assertion can be traced immediately

Planning lesson. The fact-checking matrix is the artifact worth stealing. Every assertion in the published product maps to specific documents. Any investigation, in any domain, benefits from the same table before its report is delivered, and it typically finds two or three assertions with no supporting item.

3.5 Corporate Due Diligence: A Pre-Acquisition Review Under a Deadline

(composite)

Situation. An acquirer has 21 days of exclusivity before signing a \$47M acquisition. Counsel commissions open source and records diligence on the target.

Planning. The defining constraint is the deadline, and the plan is built backwards from it. Scope: the target entity and every affiliate, its two principal facilities, its litigation and regulatory history, its key-person exposure, and the representations made in the letter of intent. Excluded: employees other than named officers, customers, and anything requiring contact with the target's staff, which would breach the process.

Objectives. Five, each mapped to a specific representation in the letter of intent, and each phrased as "determine whether the public record supports or contradicts representation N." This mapping is the planning technique that makes diligence tractable: the objectives are handed to you by the document you are testing.

Priorities. Latency-driven throughout. Court record retrieval in four jurisdictions and one county records request were initiated on day one. Entity expansion, which unlocks everything else, was completed by day two. Imagery orders placed on day three. Stable, low-latency research filled the remaining days.

Evidence strategy. Every record obtained as an official or downloadable copy, hashed, and logged with its retrieval details. Findings destined for a negotiation had to survive opposing counsel's scrutiny, so the plan required that every finding cite an item and every negative state its coverage boundary.

Documentation. A representation-by-representation workbook: one tab per representation, with supporting items, contradicting items, and a determination. This structure meant the final deliverable was assembled rather than written.

Custody. Simple but strict: single custodian, hashes at acquisition, and a production log recording every copy provided to counsel and to the acquirer, because findings used in a negotiation get

circulated and untracked copies become a problem.

Case management. A 21-day plan with daily standing check-ins, a day-14 review, and a hard day-19 stop to allow assembly. The day-14 review reprioritized after entity expansion produced a subsidiary not named in the letter of intent.

Challenges. The county records request did not arrive within the window. The plan had anticipated this and specified the fallback: report the gap, state what it would have resolved, and recommend a specific indemnity to cover the residual risk. This is a better outcome than a guess, and counsel used it directly in the negotiation.

Outcome. Two of five representations were contradicted by the public record: undisclosed litigation involving a subsidiary discovered through entity expansion, and an environmental notice with a compliance deadline falling after the planned closing. The deal proceeded at a reduced price with specific indemnities. The unresolved records gap was disclosed as a gap rather than papered over.

Planning lesson. When a deadline is fixed, plan backwards from it and identify latency before anything else. The entire diligence outcome was determined by which requests went out on day one.

3.6 Competitive Intelligence: A Standing Programme, Not a Case

(composite)

Situation. A manufacturer establishes a small competitive intelligence function. This case is included because it illustrates planning for **continuing work** rather than a bounded investigation, and the planning differences are instructive.

Element	How it differs from a case
Scope	A standing scope covering competitor set, geography, and topic areas, reviewed quarterly rather than fixed
Objectives	Standing objectives (maintain a current picture of the competitor set) plus episodic objectives raised by specific decisions
Priorities	Reset every cycle against the decisions currently in front of the business
Evidence strategy	Recurring collection against a fixed source list, with per-source cadence; the plan specifies what is collected weekly, monthly, and quarterly
Documentation	An entity file per competitor, appended continuously, with dated entries; a change log per competitor so that "what is new" is answerable
Custody	Lighter than a litigation-facing case, but still hashed and logged, because a finding may become the basis of a business decision worth millions

Element	How it differs from a case
Case management	Programme management: a source health check (is each source still returning material), a quarterly scope review, and a retirement decision for competitors no longer relevant
Ethics	The prohibited-activity statement is the most important document in the programme, because the work is continuous and the temptations recur
Challenges	Source decay (platforms change, sources close), accumulation without review, and scope creep toward individuals at competitor firms
Outcome	Over two years, three decisions were materially informed. The programme's most valuable artifact turned out to be the dated change log per competitor, because it answered "when did this start" questions that no single collection could

Planning lesson. Continuing programmes fail from accumulation, not from insufficiency. The quarterly review that retires competitors, prunes sources, and re-derives objectives is what keeps the programme useful, and it is the first thing organizations stop doing.

3.7 National Security Operations: Planning Under Compartmentation

(historical pattern, described in general terms)

Situation. Intelligence organizations plan collection and analysis under constraints that most investigators never face, and two of their practices generalize usefully.

Element	How it runs
Authority	Explicit, statutory, and documented, with the authority for each collection type distinct and separately approved
Scope	Bounded by authority as much as by intent; certain collection is unavailable regardless of value, and the plan states this
Objectives	Derived from formally endorsed requirements, with priority set centrally rather than by the collector
Priorities	Formal prioritization frameworks, reviewed at defined intervals, with resource allocation following the ranking
Evidence strategy	Multi-discipline, with the plan specifying which

Element	How it runs
	discipline addresses which requirement and how the results will be fused
Documentation	Extensive and standardized, with source characterization and confidence conventions applied uniformly
Custody	Compartmented handling, with access on a need-to-know basis, logged, and with source-protection constraints that limit what may be recorded even internally
Case management	Formal review cycles, oversight bodies, inspectors general, and post-hoc accountability that most private investigations never experience
Challenges	Compartmentation impedes fusion: the person who needs a finding may not be cleared to receive it, which is the documented cause of several historical failures
Outcome	Where it works, sustained multi-year coverage and formal accountability. Where it fails, the failure is usually dissemination or fusion rather than collection

Two practices worth generalizing.

Authority is documented per collection type, not per investigation. In private work the analogue is recording what specific authorization permits which collection activity, rather than a blanket engagement letter. It is more work and it answers the challenge question directly.

Compartmentation has a cost that must be planned for. Any investigation that restricts information internally for good reasons must also plan how findings will reach the person who needs them, or it will reproduce the classic failure in miniature.

3.8 Cross-Case Comparison

Case	Defining constraint	Planning decision that mattered most	Documentation or custody feature that saved it
Payment diversion	Log retention window	Preserving logs within four hours	Original message files with headers, not forwards
Burglary series	Inherited evidence from eleven files	Making custody reconciliation an	Reconciliation table with per-item

Case	Defining constraint	Planning decision that mattered most	Documentation or custody feature that saved it
		objective	verification status
Grant expenditure	Anonymous, unspecific complaint	Scope exclusions protecting the complainant and staff	Coverage boundaries on every negative vendor finding
Newsroom project	Publication deadline plus legal exposure	Narrowing the question deliberately	Fact-checking matrix mapping assertions to documents
Due diligence	21-day fixed deadline	Latency assessment on day one	Representation-by-representation workbook
CI programme	Continuity rather than closure	Quarterly review with retirement decisions	Dated change log per competitor
National security	Authority and compartmentation	Documenting authority per collection type	Uniform source characterization conventions

Two patterns run through all seven. **In six of the seven, the decision that determined the outcome was made during planning, before any evidence was examined.** And **in every case the documentation practice that mattered was one that preserved a limitation:** a coverage boundary, a custody exception, a records gap, a pre-investigation handling event. Documentation that only records successes is not documentation; it is marketing.

Part 4: Chapter Summary

An investigation plan is a written document produced before substantive work begins. It states what the investigation covers and excludes, what it must accomplish, in what order, using what evidence, documented how, managed by whom, and finished when. It is not a formality and not a schedule; it is what converts a concern into work that can be executed, reviewed, defended, and finished. Without scope, the effort expands without limit. Without objectives, there is effort without completion. Without priorities, the wrong work is done by the deadline. Without documentation and custody, correct findings collapse under challenge. And without any plan there is nothing to review, which is why a plan is the first thing a competent reviewer and an opposing party both ask for.

The plan is proportional, not optional. A two-day investigation gets a one-page plan in which every section appears in compressed form. Nothing is deleted. Most small investigations that go wrong go wrong because nobody spent twenty minutes.

Scope is bounded along four dimensions: subject, temporal, geographic or jurisdictional, and activity. A statement that bounds two of them is a suggestion. A complete scope section has inclusions,

exclusions, boundaries, and assumptions, and the exclusions do the most work: they prevent drift, protect the parties who are out of scope, tell the consumer what they are not buying, and tell a reviewer that an omission was a decision. Scope creep is controlled by a written change procedure whose default is stop: an out-of-scope thread is recorded, not collected, and expansion requires an approved amendment.

Objectives state what must be accomplished, measurably. The test is whether two people looking at the work would agree it is done. Each objective carries a success criterion and a completion test that a reviewer can actually perform. Every objective must be completable by a negative finding, because an objective phrased as "establish that X" pays the investigator to reach a particular result, while "determine whether X can be established" can be completed either way. Above objectives sits one strategic goal tied to the consumer's decision; below them sit tactical tasks, each citing the objective it serves.

Priorities are decided in the plan because during execution the loudest thread wins. Rank by decision impact, override for perishability, and overlay cost and latency so that slow items begin on day one. Perishable evidence is captured first regardless of rank, because online listings, profiles, logs, and camera footage disappear on their own schedules. And the plan must name what gets dropped if resources fall short, because otherwise the deadline decides.

An evidence collection plan adds four things to a collection plan: the collection method, the documentation required at the moment of collection, the legal and authorization basis, and the preservation procedure. Method matters concretely: download original files rather than printing to new ones, capture full pages with URLs and timestamps rather than crops, export structured data natively with the query recorded, and obtain original messages with headers rather than forwards. Physical evidence is preserved by seals and packaging; digital evidence is preserved by hashes. In OSINT work almost everything is digital, which makes the hash the seal and the custody log the packaging.

Documentation is substantive because reproducibility, memory, and challenge are real. The set includes the plan, contemporaneous investigator notes, an evidence inventory, custody records, a search log, capture records, a decision log, a lead register, timelines, analytic work product, communications records, and a closure record. Evidence and work product are separated. Notes are contemporaneous, dated, attributed, sequential, factual, complete on negatives, and corrected visibly rather than silently. Captures record the URL, the time with zone, the tool, and the conditions, and originals are never annotated. The best single check is the reproducibility test: hand the file to a colleague with no context and ask them to trace three findings.

Chain of custody answers one question: is this the same item, in the same state, as when it was collected? It requires a unique item ID assigned at acquisition, a hash computed before anything else, a preservation copy that is never opened, working copies for all analysis, controlled and logged access, hashing on both ends of every transfer, scheduled verification with all results recorded including successes, and authorized, dated disposal. Chains break through gaps, unlogged access, hash mismatches, late hashing, shared credentials, untracked copies, work on the original, retroactive entries, and anonymous entries. When a break occurs it is reported, not repaired: reconstructed custody entries are worse than an acknowledged gap, and a concealed break destroys the credibility of everything.

Case management runs the investigation as an organized effort through initiation, planning, execution, recurring review, analysis and production, delivery, closure, and archival. The case file has one location and a defined structure that separates preservation from working copies and evidence from work product. Tasks derive from objectives; leads arise during work and every lead receives a recorded disposition. Small investigations fail from missing documentation and large ones from missing coordination, so the plan sections that matter differ with scale. Manual and software-based management both work, and a disciplined investigator with spreadsheets beats an undisciplined one with a platform; doing it manually teaches what the platform is doing and when it is misconfigured.

Closure is a defined act. Objectives assessed, leads dispositioned, inventory reconciled, custody complete, documentation tested, report delivered and receipt confirmed, feedback recorded, exceptions disclosed, retention applied and disposal calendared. The closure memorandum should be honest about what was not accomplished, because a case that met four of six objectives and says so is more credible than one that reports only its successes.

Everything traces. Authority to scope to objectives to priorities to plan rows to evidence items to custody records to findings to the report to closure. Pick any evidence item and you should be able to state what authorized its collection, which objective it served, who collected it, when, how, its acquisition hash, every custody event since, and which finding it supports. A missing link makes that item indefensible, and every finding resting on it inherits the defect. And across all seven case studies, the decision that determined the outcome was made during planning, before any evidence was examined, while the documentation practice that saved the work was always one that preserved a limitation rather than a success.

Part 5: Key Terms

Grouped by theme. Terms carried from earlier chapters are marked and restated briefly.

5.1 Planning Foundations

Investigation plan. A written document, produced before substantive work, stating the authority, scope, objectives, priorities, evidence, documentation, custody, and management arrangements for an investigation.

Authority (in planning). The recorded basis on which an investigation is conducted: who authorized it, under what power or engagement, and when.

Tasking. The formal assignment of an investigation, recording its authority and initial requirements.

Plan amendment. An approved, versioned change to scope or objectives, as distinct from silent drift.

Version control. The practice of retaining every prior version of a plan or artifact so that changes are visible.

Revision history. The record of what changed in a plan, when, why, and who approved it.

Proportionality (of planning). The principle that plan sections compress with the size of the

investigation but are never deleted.

5.2 Scope

Scope. The boundary of the investigation: what is inside the work and what is outside it.

Subject dimension. Which entities, systems, accounts, or activities are in scope.

Temporal dimension. The period the investigation covers.

Geographic or jurisdictional dimension. The places, jurisdictions, or platforms in scope.

Activity dimension. The types of activity being examined.

Inclusions. What the investigation covers, stated specifically enough to be executed.

Exclusions. What is deliberately out of scope, and why. The most valuable and most omitted part of a scope statement.

Boundaries (method). Limits on method and access: what will not be done, which sources will not be used, what requires escalation.

Assumptions (in a plan). What the plan takes as given, each of which is a risk if wrong.

Scope creep. Gradual expansion of an investigation beyond its stated boundary without a decision to expand it.

Scope change procedure. The written process by which an out-of-scope thread is recorded rather than collected, and expansion is approved or declined.

Out-of-scope observation. A thread noted without substantive collection, recorded in the decision log pending a scope decision.

5.3 Objectives and Priorities

Objective. A statement of what the investigation must accomplish, written so that completion can be verified.

Measurability. The property of naming a deliverable, standard, or determination that can be checked.

Success criterion. What completion of an objective looks like, concretely.

Completion test. How a reviewer verifies that an objective was met.

Strategic goal. The single statement of what the investigation is for, tied to the consumer's decision.

Tactical task. An individual unit of assignable work, derived from an objective, producing a specific output.

Objective hierarchy. The structure running from strategic goal through objectives to tasks.

Negative-path completability. The property that an objective can be completed by a documented negative finding as well as by a positive one.

Prioritization method. The stated basis for ranking work: decision impact, perishability, dependency, cost and latency, risk, or external deadline.

Decision impact. How much an output would change the consumer's decision; the default primary ranking.

Perishability. The rate at which evidence disappears; grounds for overriding the impact ranking.

Perishability override. The rule that perishable evidence is captured first regardless of its impact ranking.

Latency (of collection). The interval between requesting evidence and receiving it; drives scheduling rather than ranking.

Slow item (Ch.4). A collection line with days or weeks of latency, initiated at the start of an effort.

Drop order. The written statement of what is descoped first if resources fall short.

Descoping. Formally removing an objective or task from the plan, recorded rather than abandoned.

5.4 Evidence

Evidence (Ch.5). An item bearing on whether a proposed fact is true, from a source, carrying reliability and credibility ratings.

Work product. Material the investigator produced (notes, timelines, charts, matrices), as distinct from evidence collected from the world.

Evidence collection plan. The plan section specifying, per row, what evidence is sought, from what source, by what method, under what authority, with what documentation and what preservation.

Collection method. The specific technique used to capture an item, chosen to preserve what matters about it.

Full-page capture. A screenshot or saved page covering the entire page, with the URL and timestamp visible or recorded.

Original file. The file as it exists on the source system, obtained by download rather than re-created by printing or exporting.

Native export. Structured data exported in the format the source system offers, with the query recorded.

Certified copy. An official copy of a public record, attested by its custodian.

Physical evidence. A tangible item, unique, preserved by packaging, sealing, and secure storage.

Digital evidence. Data preserved as a bit stream, verified by hash, of which perfect copies are possible.

Volatility. The tendency of evidence to change or disappear, high for remote content and system state.

Preservation. Keeping evidence in the state it was collected, and being able to demonstrate that later.

Preservation copy. A copy created at acquisition, verified, and never opened.

Working copy. A copy made from the preservation copy, used for all analysis, which may be annotated if labelled.

Hash. A cryptographic digest of a file's contents, used as a fingerprint. SHA-256 is the current standard for this work.

Hash verification. Recomputing a hash and comparing it to the recorded value.

Hash mismatch. A computed hash differing from the recorded value; must be reported, never overwritten.

Write protection. Storage configured so that files cannot be altered.

Retention schedule. The defined period for which material is kept, and the disposal date.

Disposal. Authorized destruction or return of evidence, with the method and date recorded.

5.5 Documentation

Contemporaneous record. A record made at the time of the action rather than reconstructed later.

Late entry. A note written after the fact, marked as such with the date of entry.

Investigator notes. The sequential, dated, attributed record of what was done and observed.

Struck-through correction. An error corrected visibly, dated and initialled, rather than deleted.

Evidence log / inventory. The register of every evidence item with its source, acquisition details, hash, and custody reference.

Search log. The record of every query: system, terms, filters, date range, result count, and timestamp.

Capture record. The record accompanying each screenshot or page capture: URL, timestamp, tool and version, conditions.

Decision log (Ch.4). The record of significant investigative decisions, the alternatives, and the reasons.

Lead register. The record of every lead, its source, priority, status, and disposition.

Lead disposition. The recorded outcome of a lead: worked, resolved, closed as unproductive, out of scope, or referred.

Communications record. The record of requests, approvals, instructions, and exchanges with the consumer or third parties.

Closure memorandum. The document recording what was accomplished, what was not, what remains open, and any exceptions.

Reproducibility test. Handing the case file to someone with no context and asking them to trace findings to their sources.

Standardized template. A fixed form for a recurring record type, which makes review and comparison possible at volume.

Fact-checking matrix. A table mapping every assertion in a product to the specific items supporting it.

5.6 Chain of Custody

Chain of custody. The documented, unbroken record of everyone who had control of an item from acquisition to disposition, with every transfer, storage change, and verification recorded.

Item ID. A unique identifier assigned at acquisition and never reused.

Acquisition entry. The founding custody entry, recording what was collected, from where, by whom, when, how, and its hash.

Custody event. Any acquisition, transfer, access, verification, duplication, storage change, or disposal.

Transfer. The movement of an item between people or systems, hashed and recorded at both ends.

Custody gap. A period during which an item's location and control are not accounted for.

Custody break. Any defect that prevents the record from establishing continuity, including gaps, unlogged access, mismatches, and retroactive entries.

Unlogged access. Any interaction with an item that is not recorded, including opening a preservation copy.

Retroactive entry. A custody entry written after the event, which records recollection rather than the event.

Evidence custodian. The designated person responsible for holding evidence and maintaining custody records, in larger investigations distinct from the investigators.

Production log. The record of every copy of evidence provided to any recipient.

Scheduled verification. Periodic re-hashing on a defined cadence, with all results recorded.

5.7 Case Management

Case management. The discipline of running an investigation as an organized effort: tasks, leads, case file, coordination, review, and closure.

Case number. The unique identifier for the investigation, referenced by every artifact.

Case file. The single structured store containing all administration, planning, evidence, custody, documentation, analysis, products, and closure records.

Case lifecycle. Initiation, planning, execution, review, analysis and production, delivery, closure, and archival.

Task register. The record of tasks with owners, dependencies, due dates, status, and outputs.

Series case. A case consolidating several previously separate matters under one investigation.

Custody reconciliation. Verifying that every item logged is present and every item present is logged, with continuity established.

Deconfliction. Coordinating among investigators so that collection and contact are not duplicated.

Review point. A scheduled checkpoint assessing progress against objectives, evidence integrity, documentation, and scope.

Closure criteria. The stated conditions that must be met before a case is closed.

Archival. Sealing the case file, applying the retention period, and calendaring disposal.

Standing programme. Continuing work organized around standing scope and objectives with a review cadence, as distinct from a bounded case.

Source health check. A periodic confirmation that each recurring source is still returning material.

Retirement (of scope or a source). The deliberate, documented ending of coverage that no longer serves a decision.

Part 6: Knowledge Assessment

6.1 Short Answer (20 questions)

Answer in three to five sentences unless otherwise noted.

1. Define an investigation plan and name six of its required sections.
2. Explain the four failures that occur without scope, objectives, priorities, and documentation respectively, and name the fifth failure that sits above all of them.
3. Explain what proportional planning means and why no section is ever deleted for a small investigation.
4. Name the four dimensions of scope and give an example of each.
5. Explain why exclusions are the most valuable part of a scope statement, and give two things they accomplish.
6. Define scope creep, name three causes, and describe the control that prevents it.
7. Explain why the default in a scope change procedure must be stop.
8. Explain what makes an objective measurable, and give one unmeasurable objective with a measurable rewrite.
9. Explain why every objective must be completable by a negative finding.
10. Distinguish a strategic goal, an objective, and a tactical task on at least four dimensions.
11. Explain the difference between a success criterion and a completion test, and why the second matters.
12. Name four prioritization methods and state which is the default and why.

13. Explain the perishability override and give three examples of perishable evidence with their typical windows.
14. Explain why a plan must name a drop order.
15. Name the four elements an evidence collection plan adds beyond a list of sources.
16. Explain why a document should be downloaded rather than printed to a new PDF.
17. Compare physical and digital evidence on integrity mechanism, alteration visibility, and working practice.
18. Explain what a hash establishes and name three things it does not establish.
19. Explain Rule 7 and give three substantive reasons documentation matters beyond procedure.
20. Explain what chain of custody is for, name four ways it breaks, and describe the correct response to a break.

6.2 Multiple Choice (15 questions)

1. An investigation plan is best described as: a) A schedule for the work b) A written document produced before substantive work stating scope, objectives, priorities, and procedures c) A summary written at the end for the file d) A legal disclaimer
2. Which scope element is most often omitted and does the most work? a) Inclusions b) Exclusions c) The temporal boundary d) Assumptions
3. A scope statement bounding only subject and activity is: a) Complete b) Adequate for small investigations c) Unbounded along the temporal and geographic edges d) Preferable, because it allows flexibility
4. In a scope change procedure, the correct default when an out-of-scope thread appears is: a) Collect it now and seek approval afterward b) Stop, record the observation without substantive collection, and seek a decision c) Collect it if it is cheap d) Ignore it entirely and record nothing
5. "Understand the counterfeiting problem" fails as an objective primarily because it: a) Is too long b) Cannot be verified as complete by two people c) Is outside scope d) Presupposes wrongdoing
6. An objective phrased "establish that the vendor overbilled" is defective because it: a) Is unmeasurable b) Can only be completed by a positive finding, which pressures the result c) Names a vendor d) Lacks a deadline
7. Which statement about tactical tasks is correct? a) They change rarely and only by amendment b) They are owned by the consumer c) Each should cite the objective it serves d) They are measured by whether the decision was supported
8. The default primary basis for prioritizing investigative work is: a) Cost b) Perishability c) Decision impact d) Investigator interest
9. A high-impact records search with three-week latency and a low-impact listing capture that will be deleted this week should be sequenced as: a) Records search first, because impact ranks higher b)

Listing capture first, and the records request initiated immediately as a slow item c) Both deferred until the plan is approved d) Whichever is more convenient

10. Downloading a filing rather than printing it to a new PDF matters because printing: a) Reduces image quality b) Produces a different file with different metadata and a different hash c) Is slower d) Is prohibited by most terms of service

11. A matching SHA-256 hash establishes that: a) The content of the file is truthful b) The source was authentic c) The file is bit-for-bit identical to the file hashed at acquisition d) The collection was authorized

12. Annotating an original capture with an arrow is improper because it: a) Looks unprofessional b) Creates a new file, changes the hash, and destroys the original c) Is unnecessary d) Violates the individuals policy

13. On discovering a gap in a custody record, the correct action is to: a) Reconstruct the missing entries from memory b) Report it, determine its extent, assess the impact on affected findings, and state it in the report c) Remove the affected item from the inventory d) Note it internally but omit it from the report

14. In small versus large investigations, the dominant risks are respectively: a) Coordination and documentation b) Documentation and coordination c) Cost and time d) Legal exposure in both

15. A case is properly closed when, among other conditions: a) The report has been sent b) Every objective is assessed, every lead is dispositioned, the inventory reconciles, and retention is applied c) The consumer stops asking questions d) The deadline passes

6.3 True or False (10 questions)

Mark true or false. For each false statement, write one sentence explaining what makes it false.

1. A two-day investigation does not need a written plan.
2. Exclusions in a scope statement tell a reviewer that an omission was a decision rather than an oversight.
3. Priorities should be set during execution, when the relative importance of threads becomes clear.
4. Perishable evidence should be captured ahead of more important but stable evidence.
5. A hash computed a week after collection proves the file was unaltered from the moment of collection.
6. The preservation copy of a digital item should be used for analysis to avoid creating unnecessary copies.
7. Successful hash verifications do not need to be recorded, only failures.
8. Recording searches that returned nothing is unnecessary detail.
9. A lead that is no longer mentioned in the case file has been closed.

10. A disciplined investigator using spreadsheets can produce a better case file than an undisciplined investigator using a case management platform.
-

Part 7: Critical Thinking Exercises

Each group targets one planning skill. Produce the artifact the exercise names.

Defining investigation scope

7.1 A hospital system's compliance officer says: "We think someone is selling patient appointment slots. Look into it." Write a scope statement with all four dimensions, at least five exclusions with reasons, at least four method boundaries, and at least three assumptions. Then identify the one exclusion you consider most important and explain why.

7.2 Two weeks into an investigation scoped to a company's marketing claims, you notice that the company's building permits appear irregular. Apply the scope change procedure in writing: what you record, what you do not do, what the amendment would say, and what it would cost and displace.

7.3 Critique this scope statement and rewrite it: "This investigation will examine all activity relating to Meridian Freight and its associates over recent years using all available sources to determine who is responsible."

Developing measurable objectives

7.4 Convert each into a measurable objective with a success criterion and a completion test: (a) "Find out who is behind the accounts"; (b) "Assess the vendor's financial health"; (c) "Understand how the document leaked"; (d) "Verify the company's claims."

7.5 Write a strategic goal, four objectives, and six tactical tasks for one objective, for an investigation of your choosing. Then explain what would happen at closure if the objectives were missing and only the tasks existed.

7.6 An objective reads: "Document the extent of the fraud." Explain the two distinct defects in this objective and rewrite it.

Prioritizing investigative activities

7.7 You have eleven tasks, three weeks, and one investigator. Four tasks concern evidence with a two-week retention window; two require records requests with three-week latency; three are high-impact and stable; two are low-impact and interesting. Produce a priority matrix with a stated method, a schedule, and a drop order, and defend the sequence.

7.8 Halfway through, a finding raises the impact of a task you ranked ninth. Describe what you do, what you record, and whether this is a reprioritization or a plan amendment.

7.9 Your consumer says every objective is equally important. Write the response that gets you a usable ranking, and explain why accepting "all equally important" is a planning failure rather than a courtesy.

Designing evidence collection strategies

7.10 For an investigation into online resale of counterfeit goods, write six evidence collection plan rows in full, including method, documentation required at collection, legal basis, perishability, and preservation procedure. At least two rows must concern perishable evidence and at least one must be a slow item.

7.11 A client offers to give you screenshots they took themselves of the listings, to save you time. Assess this offer: what you accept, what you do not, what you record, and why the client's screenshots are not a substitute for your own captures.

7.12 You need a public record that exists only on a terminal at a county office, with no download available. Describe your collection method, your documentation, and how you would establish integrity for an item you cannot hash at the source.

Creating documentation plans

7.13 Write a documentation plan for a three-week, single-investigator OSINT investigation: which artifacts, in what format, stored where, with what naming convention, and reviewed when.

7.14 Take one of your own recent pieces of research, coursework or otherwise. Apply the reproducibility test: could a stranger trace three findings to their sources using only what you wrote down? Report honestly what they could and could not do, and name the two artifacts whose absence caused the most difficulty.

7.15 An investigator's notes for a full day read: "Did searches on the target. Found some listings. Will follow up." Rewrite this as a professional note entry set, inventing plausible specifics, and list every field the original omitted.

Maintaining chain of custody

7.16 You acquire 14 web captures, 3 downloaded PDFs, and one client-provided email file. Design the custody arrangement: item IDs, hashing, preservation and working copies, storage, access, transfer procedure, and verification schedule.

7.17 During verification you find that one item's computed hash does not match the recorded value. Write the entries you would make, the notifications you would send, the determination process you would follow, and the sentence you would put in the final report.

7.18 A colleague suggests that since digital copies are identical, only one copy is needed and hashing is unnecessary for open source material that anyone could re-collect. Write the response, addressing re-collectability, alteration, and what a challenge would target.

Managing cases from beginning to end

7.19 Design a case file structure for a six-week investigation with two investigators, and explain the two structural decisions you consider most important.

7.20 You inherit a partially worked case with 40 evidence items, no plan, and notes in three formats.

Write your first-week plan: what you do first, what you reconcile, what you may not be able to recover, and what you tell the consumer.

7.21 Write a closure memorandum for an investigation that met three of five objectives, has two open leads, one custody exception, and an unresolved records request. Then explain why an honest closure memorandum is more credible than one reporting only successes.

Part 8: Hands-On Lab Exercises

General Lab Guidance

All five labs run on one fictional investigation, from plan through closure. Lab 1 produces the plan, Lab 2 the evidence collection plan, Lab 3 the documentation set, Lab 4 the custody records, and Lab 5 the case management and closure package. Do them in order and keep every artifact; Lab 5 requires all of them.

Everything is fictional. Larkspur Outfitters, every seller, every person, and every address are invented. Nothing here describes a real company or product.

Rules 1 through 7 apply. Rule 7 in particular: if it is not written down, it did not happen, and the rubrics grade accordingly.

Rubric scale. 0 to 4 per criterion: 4 professional, 3 competent, 2 developing, 1 attempted, 0 missing or fundamentally incorrect.

The Chained Case: Larkspur Outfitters

Fictional in all respects.

The client. Larkspur Outfitters is a mid-sized outdoor equipment manufacturer, roughly \$90M in annual revenue, selling through 140 authorized dealers and its own web store. Its marks are registered.

The trigger. Two things arrived in the same week.

First, Larkspur's customer service team received nine complaints in twelve days about products that failed in ways Larkspur's products do not fail, from customers who had bought online from sellers Larkspur does not recognize.

Second, a Larkspur product engineer found a PDF of an internal product specification document, marked "Larkspur Internal," available on a public file-sharing site. The document covers a product that has not yet been released.

The consumer. A. Whitfield, Larkspur's general counsel, who must decide by 30 June (six weeks from initiation) among four courses of action:

1. Platform enforcement only: submit takedown requests to the marketplaces.

2. Platform enforcement plus civil action against identified sellers.
3. Referral to law enforcement, if the evidence supports it.
4. Internal review of the document exposure, handled separately by counsel.

Your role. You are the sole investigator, engaged by counsel. Your engagement letter authorizes open source investigation and records research. It does not authorize test purchases, contact with sellers, or any examination of Larkspur personnel.

Known facts at initiation.

#	Fact	Source
K1	Nine customer complaints in twelve days, each naming an online seller	Client records
K2	Four distinct seller names appear across the nine complaints	Client records
K3	Larkspur has 140 authorized dealers; none of the four seller names is on the list	Client records
K4	The specification document concerns an unreleased product with a planned launch in September	Client records
K5	The document was found on a public file-sharing site by a Larkspur engineer, who took a screenshot and sent it by email to the engineering director, who forwarded it to counsel	Client records
K6	Larkspur's marks are registered and the registration numbers are available	Public records
K7	Larkspur's product photography is copyrighted and its image files are distinctive	Client records
K8	The document is 14 pages and includes a footer with a document control number	Client records

Lab 1: The Investigation Plan

Time required: 2.5 to 3 hours **Format:** Individual **Difficulty:** Intermediate

Objectives

1. Write a scope statement bounded on all four dimensions with substantive exclusions.
2. Write measurable objectives with success criteria and completion tests.
3. Prioritize tasks using a stated method with a perishability override and a drop order.
4. Produce a complete, approvable investigation plan.

Prerequisites

- Sections 1.1 through 1.4 and comparison tables 2.1, 2.2, and 2.9.
- No collection. This is a planning lab, and beginning to search is a deduction.

Steps

Step 1 (15 min). Authority and initiation. Write the plan's opening section: who authorized the investigation, on what basis, when, what the engagement permits and does not permit, and the case number you assign. State the strategic goal in one sentence, tied to Whitfield's 30 June decision.

Step 2 (35 min). Scope statement on Worksheet 1A. All four dimensions. Minimum six exclusions, each with a reason. Minimum five method boundaries. Minimum four assumptions. Note that K5 describes a pre-investigation handling chain and that the document exposure may implicate Larkspur personnel, which the engagement excludes.

Step 3 (40 min). Objectives on Worksheet 1B. Four to six objectives, each with a rationale, a measurable statement, a success criterion, a completion test, dependencies, an owner, and a deadline. Every objective must be completable by a negative finding, and the rubric checks this specifically.

Step 4 (25 min). Task derivation on Worksheet 1C. For each objective, three to six tactical tasks, each with an output and an estimated duration. Minimum eighteen tasks total. Every task cites its objective ID.

Step 5 (30 min). Priorities on Worksheet 1D. Score every task on decision impact, perishability, and latency. Produce a composite ranking, a three-week schedule for the first phase, and an explicit drop order naming what is descope first, second, and third if you lose a week.

Step 6 (15 min). Scope change procedure and risks. Write the procedure, with a default of stop. Then list at least five risks or constraints (deadline, perishability, platform terms, jurisdictional limits, the possibility that the document exposure is internal) and your mitigation for each.

Step 7 (15 min). Closure criteria and approvals. State what will be true when this investigation is complete. Add an approval block and a revision history table with version 1.0.

Step 8 (10 min). Self-review against the plan review checklist in Section 1.9.2. Record any item you could not satisfy and why.

Worksheet 1A: Scope Statement

Element	Content
Subject dimension	
Temporal boundary	
Geographic / jurisdictional boundary	
Activity boundary	
Inclusions (a, b, c...)	
Exclusions (minimum 6, each with a reason)	
Method boundaries (minimum 5)	
Assumptions (minimum 4, each a registered risk)	

Worksheet 1B: Objectives Register

Obj ID	Statement (measurable)	Rationale	Success criterion	Completion test	Dependencies	Owner	Deadline	Negative-path completable?

Worksheet 1C: Task Register

Task ID	Obj ID	Description	Output	Duration	Owner

Worksheet 1D: Priority Matrix

Task ID	Decision impact	Perishability	Latency	Composite rank	Week	Drop order

Expected Outcomes

- A scope statement bounding all four dimensions. Students frequently leave the subject dimension unbounded by writing "all activity relating to Larkspur," which would include the client's own staff and is the error the K5 facts are designed to expose.
- **The critical exclusion:** the document exposure investigation must be bounded to what public sources establish, with any indication of internal involvement referred to counsel rather than pursued, because the engagement does not authorize examining Larkspur personnel. Strong submissions state this explicitly and explain the referral pathway.
- Objectives that are completable by negative findings. "Identify the sellers" fails; "determine whether a business entity can be identified from public records for each seller account,

recording either the identification with supporting records or a negative finding with its coverage boundary" passes.

- **The perishability finding:** listings and seller profiles are the most perishable evidence in the case and must rank first regardless of impact. The file-sharing site posting is equally perishable and equally urgent. Students who rank registry research first have missed the override.
- At least one slow item identified: county records, certified copies of registrations, or any request with multi-day latency.
- A drop order that names specific descope tasks rather than saying "we would prioritize."
- A scope change procedure whose default is stop.

Discussion Questions

1. Which exclusion did you find hardest to write, and what pressure would push against it during execution?
2. K5 describes the document passing through three people before reaching you. What does that mean for your evidence plan, and is it a problem you can fix?
3. Your highest-priority tasks are probably not your highest-impact tasks. Explain that to Whitfield in two sentences.
4. What is the first thing you would descope, and what would Larkspur lose by that?

Grading Rubric: Lab 1 (56 points)

Criterion	Weight	4 points looks like
Authority and strategic goal	x2	Authorization recorded; goal tied to the 30 June decision
Scope: four dimensions bounded	x3	All four specific and testable
Exclusions	x3	Six-plus, each with a reason; personnel exclusion and referral pathway present
Method boundaries	x2	Five-plus, specific to this engagement
Assumptions	x2	Four-plus, each a genuine risk
Objectives measurable	x3	Success criterion and completion test on each
Negative-path completability	x3	Every objective completable either way
Task derivation	x2	Eighteen-plus, each citing an objective, each with an output
Prioritization method stated	x2	Method explicit; composite

Criterion	Weight	4 points looks like
		ranking defensible
Perishability override applied	x3	Listings and posting ranked first
Slow items scheduled first	x2	Identified and placed in week one
Drop order	x2	Specific tasks named in order
Scope change procedure	x2	Default stop
Closure criteria and version control	x1	Stated; approval block and revision history present

Total possible: 128 weighted. Passing: 88.

Lab 2: The Evidence Collection Plan

Time required: 2.5 to 3 hours **Format:** Individual **Difficulty:** Intermediate

Objectives

1. Build an evidence collection plan with method, documentation, legal basis, and preservation per row.
2. Select collection methods appropriate to each evidence type.
3. Apply the four tests per row before collection.
4. Specify preservation procedures including hashing, storage, and verification.

Prerequisites

- Section 1.5 and comparison tables 2.3 and 2.5.
- Lab 1 complete. Bring the objectives and priority matrix.

Steps

Step 1 (20 min). Evidence identification on Worksheet 2A. For each objective from Lab 1, list every distinct evidence item or class you will need. Minimum twenty items or classes. For each: the evidence type from table 2.3, and the objective it serves.

Step 2 (70 min). Build the plan on Worksheet 2B. One row per item or class, with every field from Section 1.5.4. Requirements for full credit:

- Minimum eighteen rows.
- At least six Chapter 3 source categories represented.
- Collection method specified for every row, chosen from table 2.3 and appropriate to the type.

- Documentation required at collection specified per row, as actual field lists rather than "document it."
- Legal basis per row: the four tests result, plus any authorization required.
- Perishability rating per row.
- Preservation procedure per row: hash algorithm, storage location, copy structure, verification cadence.
- Expected coverage boundary written in advance for every row.
- At least two rows requiring authorization you do not currently have, marked as blocked pending authorization.

Step 3 (20 min). Four tests audit on Worksheet 2C. Apply all four tests to every row. At least two rows should fail or require narrowing; the test purchase and any seller-contact row are the obvious candidates. Record the disposition of each.

Step 4 (25 min). Preservation architecture on Worksheet 2D. Specify: the storage structure, the preservation and working copy arrangement, the hashing procedure and algorithm, the access control, the backup arrangement, the verification schedule, and the retention and disposal plan. Address the specific problem that you are a sole investigator with no evidence custodian, and name the control weakness that creates.

Step 5 (20 min). Handling procedures on Worksheet 2E. Write the procedure for each of these situations, before they arise: (a) a listing you captured is deleted before the report; (b) you encounter a listing selling something outside scope; (c) the client offers you their own screenshots; (d) you find personal information about a seller that is outside scope; (e) a platform's terms prohibit the collection method you planned; (f) you encounter material that is not lawfully public.

Step 6 (15 min). Self-review. Confirm every row has all four added elements. Record any row where you could not specify a preservation procedure and why.

Worksheet 2A: Evidence Identification

#	Evidence item or class	Type (table 2.3)	Objective served
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Worksheet 2B: Evidence Collection Plan

Row ID	Obj ID	Evidence sought	Category	Source / system	Collection method	Perishability	Legal basis / authorization	Documentation required at collection	Preservation procedure	Expected coverage boundary	Status
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Worksheet 2C: Four Tests Audit

Row ID	Access	Identity	Protection	Purpose/ harm	Verdict	Disposition
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Worksheet 2D: Preservation Architecture

Element	Specification
Storage structure	
Preservation copy arrangement	
Working copy arrangement	
Hash algorithm and procedure	
Access control	
Backup	
Verification schedule	
Retention and disposal	
Sole-investigator control weakness and mitigation	

Worksheet 2E: Handling Procedures

Situation	Procedure	Recorded where
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Expected Outcomes

- Eighteen-plus rows across six-plus categories with all four added elements on every row.
- **Method appropriateness is the heavily weighted criterion.** Listings require full-page capture plus independent archive; the specification PDF requires downloading the original file, not printing; registry results require native export with the query recorded; the client-provided email requires the original file with headers rather than the forward described in K5.
- **The K5 problem must be addressed.** The document reached you through a screenshot, an email, and a forward. The correct plan requires obtaining the original PDF from the file-sharing site directly (if still available), recording the pre-investigation handling chain as a documented limitation, and treating the engineer's screenshot as a separate item with its own provenance rather than as the document.
- At least two rows blocked pending authorization: a test purchase and any seller contact. Students who plan a test purchase without flagging the authorization requirement have missed the engagement limit.
- Preservation procedures specifying SHA-256, a preservation copy that is never opened, working

copies, and a verification cadence.

- Coverage boundaries written in advance, including for registry searches ("this state only; other states unchecked").
- The sole-investigator weakness acknowledged: no separation between investigator and custodian, which means no independent check on custody, and a mitigation such as scheduled self-verification with recorded results and a peer review of the custody log at closure.
- Handling procedure (f) must state that unlawfully public material is not collected, that the encounter is recorded, and that no substantive collection occurs.

Discussion Questions

1. Which collection method did you initially get wrong, and what would have been lost?
2. K5 means the document you were given is not the document as it existed on the site. Explain the difference and what you can still establish.
3. You are both investigator and custodian. What can a challenger say about your custody record that they could not say if the roles were separated?
4. Which handling procedure do you expect to actually need, and why is writing it now better than deciding then?

Grading Rubric: Lab 2 (64 points)

Criterion	Weight	4 points looks like
Evidence identification breadth	x2	Twenty-plus items or classes, typed, mapped to objectives
Plan volume and category breadth	x2	Eighteen-plus rows, six-plus categories
Collection method appropriateness	x4	Correct method per type; originals downloaded, captures full-page, exports native
K5 handling	x3	Original sought; pre-investigation chain documented; screenshot treated as a separate item
Documentation required at collection	x3	Actual field lists per row, not generic
Legal basis per row	x3	Four tests result plus authorization status
Authorization-blocked rows identified	x2	Test purchase and contact flagged as blocked
Perishability per row	x2	Rated and consistent with Lab 1 priorities

Criterion	Weight	4 points looks like
Preservation procedure per row	x3	Algorithm, copies, storage, verification
Coverage boundaries in advance	x3	Present and realistic on every row
Preservation architecture	x2	Complete; sole-investigator weakness named with mitigation
Handling procedures	x3	All six written; (f) correctly requires no collection

Total possible: 128 weighted. Passing: 88.

Lab 3: Documenting a Simulated Investigation

Time required: 2.5 to 3 hours **Format:** Individual **Difficulty:** Intermediate

Objectives

1. Produce investigator notes, a search log, capture records, an evidence log, a timeline, and decision records from a simulated day of activity.
2. Apply the documentation standard, including negatives and visible corrections.
3. Separate observation from conclusion, and evidence from work product.
4. Pass the reproducibility test.

Prerequisites

- Section 1.6 and comparison table 2.4.
- Labs 1 and 2 complete.

The Simulated Activity Feed

You performed the following activities. Document them. Times are your local time zone; use it consistently and state it.

#	What happened
A1	09:05. Opened the case file. Reviewed the plan. Confirmed the objectives and today's task list (tasks 1.2, 1.3, 2.1 from your Lab 1 register, or the nearest equivalents).
A2	09:18. Searched Platform A using the term set "Larkspur," "Larkspur Outfitters," and two product model names, filtered to active listings

#	What happened
	shipping to the United States. Result count: 26.
A3	09:31 to 10:40. Captured 26 listings as full-page captures. Two listings failed to capture fully on the first attempt because the page loaded additional content on scroll; recaptured both using sequential captures.
A4	10:44. Hashed all 26 captures using SHA-256. Recorded hashes. Created preservation copies.
A5	10:58. Noticed that four of the 26 listings use the same product photograph, which appears to be Larkspur's own copyrighted product image.
A6	11:10. Searched Platform B using the same term set. Result count: 9. Captured all 9.
A7	11:35. Searched Platform C using the same term set. Result count: 0. Confirmed the search returned no results by repeating with a broader single-term query, which returned 4 results, none offering Larkspur-marked goods.
A8	11:50. While reviewing Platform B results, encountered a listing offering a competitor's product with what appeared to be a falsified safety certification. Not within scope.
A9	12:20. Captured seller profile pages for all distinct sellers across Platforms A and B. Eleven distinct seller accounts identified.
A10	13:15. Extracted seller-declared business names and locations from the profiles. Four profiles declared business names; seven declared none.
A11	13:40. Searched the state corporate registry for the four declared business names, plus six name variants. Two entities located. Two not located under any variant tried.
A12	14:20. Downloaded the two located entities' registry records as official PDF copies. Hashed and logged.
A13	14:35. Realised that in the notes entry written at 11:10 you had recorded the Platform B result count as 8, when the correct count was 9.
A14	14:50. Attempted to re-access one Platform A listing captured at 09:47 (your item for that capture). The listing now returns a "no longer available" page. Preservation copy retained.

#	What happened
A15	15:20. Decided not to pursue the seven sellers with no declared business name through further identity research at this stage, on the basis that the available public identifiers were insufficient and the effort would displace higher-priority tasks.
A16	15:45. Sent a written request to the client for the registration numbers of Larkspur's marks and for a copy of the authorized dealer list, both needed for objectives 1 and 2.
A17	16:10. Built a working timeline of the day's collection. Noted that the four listings sharing a product image (A5) were all created, per their profile pages, within an 11-day window.
A18	16:30. Verified all hashes recorded today. All matched. Recorded the verification.

Steps

Step 1 (55 min). Investigator notes on Worksheet 3A. Produce a numbered, dated, timed, attributed note entry for every activity. Requirements: contemporaneous format, observation separated from conclusion, negatives recorded with coverage boundaries, and the A13 error handled as a visible struck-through correction with the correction date rather than a silent fix.

Step 2 (30 min). Search log on Worksheet 3B. One row per search: system, exact terms, filters, date range, result count, timestamp, and the searcher. A7 requires two rows and the negative must be recorded with its coverage boundary.

Step 3 (30 min). Capture records on Worksheet 3C. One row per capture class (you need not write 46 individual rows; group by platform and note the item ID ranges), recording URL handling, timestamp, tool and version, capture method, and the A3 recapture issue.

Step 4 (20 min). Evidence log on Worksheet 3D. Item IDs, description, source, acquisition details, hash placeholder, preservation location, and objective served. Note that this is the evidence inventory and that your timeline from A17 is work product and does not belong in it.

Step 5 (20 min). Decision records on Worksheet 3E. A minimum of four decisions from the feed, each with the decision, the alternatives, the reason, and what would change it. A8, A14, A15, and A16 are the obvious candidates.

Step 6 (15 min). Timeline on Worksheet 3F. The day's collection, plus the A17 finding about the 11-day account creation window, with precision markers and entry types from Chapter 5.

Step 7 (15 min). Reproducibility test. Exchange with a partner, or self-test after a break using a written checklist. Pick three of your entries and confirm that a stranger could determine the source, time, method, and integrity status. Record what failed.

Worksheet 3A: Investigator Notes

Entry #	Date/time (TZ)	Initials	Entry (observation)	Conclusion, if any, labelled	Corrections
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Worksheet 3B: Search Log

SL ID	Date/time	System	Exact terms	Filters	Range	Result count	Coverage boundary	Searcher
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Worksheet 3C: Capture Records

CR ID	Item IDs	URL handling	Date/time	Tool and version	Method	Issues	Hash recorded?
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Worksheet 3D: Evidence Log

Item ID	Description	Category	Source	Acquired (date/time, method, by whom)	Hash (SHA-256)	Preservation location	Objective
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Worksheet 3E: Decision Records

DR ID	Date/time	Decision	Alternatives	Reason	What would change it
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Worksheet 3F: Timeline

Date/time	Precision	Type	Event	Item IDs	Flags
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Expected Outcomes

- Eighteen note entries, timed, attributed, contemporaneous in format.
- **A5 must be recorded as an observation with a labelled conclusion**, not as a fact. "Four listings display the same product photograph; the photograph appears visually consistent with Larkspur's product imagery" is the observation. Whether it is Larkspur's copyrighted image is a determination requiring comparison against the client's originals, which you have requested at A16 and do not yet have.
- **A7 is the negative that most students under-record.** The correct entry states the system, the exact terms, the filters, the zero result, the confirming broader search, its result count, and the

coverage boundary: this platform, these terms, this date, active listings only.

- **A13 must be a visible correction.** A struck-through original with the corrected value, the correction date, and initials. A silently fixed note loses the points and, in the discussion, students should recognize why.
- **A14 is a significant entry.** A listing captured this morning is gone this afternoon. The note records the attempted re-access, the result, the retained preservation copy, and explicitly states that the cause of removal is not determinable. It also validates the perishability priority from Lab 1, which strong submissions note.
- **A8 must be recorded and not pursued.** An out-of-scope observation, logged in the decision record with what was seen and where, no substantive collection, and a note that it may warrant referral at closure.
- **A15 is a descoping decision** and must be recorded as such with its reason, not simply omitted.
- The timeline must exclude nothing and must mark the A17 finding's precision correctly: account creation dates come from profile pages, which is reported information, and the 11-day window is exact only if the profiles state exact dates.
- The evidence log must not contain the timeline or any analytic product.

Discussion Questions

1. What did the reproducibility test reveal that you thought was adequately documented?
2. A14 shows a listing disappearing within hours. What does that validate about your Lab 1 priorities, and what would have happened if you had scheduled captures for week three?
3. Explain why the A13 correction is handled visibly, and what a silently corrected note does to the credibility of an entire file.
4. A5 is tempting to write as a finding. Write it as a finding, then as an observation, and explain the difference in what each commits you to.
5. Which of your four decision records would matter most if this investigation were challenged?

Grading Rubric: Lab 3 (72 points)

Criterion	Weight	4 points looks like
Note completeness	x2	All eighteen activities, timed, attributed
Contemporaneous format	x2	Reads as written at the time, not summarized
Observation and conclusion separated	x4	Consistent, especially at A5 and A17
Negatives recorded with boundaries	x4	A7 fully documented; other negatives present

Criterion	Weight	4 points looks like
Visible correction at A13	x3	Struck through, dated, initialled
Perishability event at A14	x3	Attempted re-access, result, preservation copy, cause undetermined
Out-of-scope handling at A8	x3	Recorded, not pursued, referral noted
Descoping decision at A15	x2	Recorded with reason
Search log quality	x3	Exact terms, filters, counts, boundaries
Capture records quality	x2	URL, time, tool, method, recapture issue noted
Evidence log discipline	x3	No work product in the inventory; all fields present
Decision records	x2	Four-plus, with alternatives and reasons
Timeline precision and types	x2	Correct precision on profile-derived dates
Reproducibility test	x2	Performed and failures honestly recorded

Total possible: 148 weighted. Passing: 102.

Lab 4: Chain of Custody for Digital Evidence

Time required: 2.5 to 3 hours **Format:** Individual **Difficulty:** Advanced

Objectives

1. Acquire, hash, and log a set of digital evidence items.
2. Build a complete chain of custody record covering acquisition, storage, transfer, verification, and preservation.
3. Reconcile an inventory against a provided manifest and detect discrepancies.
4. Identify a hash mismatch and report it correctly rather than resolving it silently.
5. Recognize common handling errors in a provided custody narrative.

Prerequisites

- Sections 1.5.5, 1.7, and comparison tables 2.5 and 2.6.

- Labs 1 through 3 complete.
- The **Lab 4 evidence set** provided alongside this workbook.

8.4.1 The Evidence Set

The provided folder contains ten simulated evidence files. All content is fictional.

File	Simulated origin
E-001_platformA_listing_cedarflat s.png	Full-page capture of a Platform A listing
E-003_platformA_listing_northbend .png	Full-page capture of a Platform A listing
E-005_platformB_listing_cedarflat s.png	Full-page capture of a Platform B listing
E-012_platformA_profile_cedarflat s.png	Seller profile capture
E-014_platformB_profile_brightfor k.png	Seller profile capture
E-021_filestash_page.png	Capture of the file-sharing page hosting the specification document
E-022_L0-SPEC-4412-RidgelineTent- rev3.pdf	The specification document, downloaded as the original file
E-031_registry_cedarflats_trading .pdf	Official copy of a corporate registry record
E-040_listing_export.csv	Native export of listing search results
E-051_client_email_ferreira.eml	Client-provided message file with full headers

8.4.2 Computing Hashes

Use whichever is available. All three are built in and nothing needs to be installed.

System	Command
Windows (PowerShell)	<code>Get-FileHash -Algorithm SHA256 . \filename</code>
Windows (Command Prompt)	<code>certutil -hashfile filename SHA256</code>
macOS	<code>shasum -a 256 filename</code>
Linux	<code>sha256sum filename</code>

To hash every file in the folder at once: PowerShell `Get-FileHash -Algorithm SHA256 *`; macOS `shasum -a 256 *`; Linux `sha256sum *`.

Record the full 64-character hash. Truncating it defeats the purpose, and a rubric criterion checks for it.

8.4.3 The Provided Manifest

This manifest was supplied to you by a previous investigator along with the evidence set. Treat it as the document of record you must reconcile against. It contains discrepancies. Finding and correctly reporting them is the point of this lab.

Item ID	File name	Recorded SHA-256	Recorded acquisition
E-001	E-001_platformA_listing_cedarflats.png	da55f9d2fffd1df482c6da208e012985f3b8d257a7848f586fee2a9f82613ffb2	12 May 09:47 CDT, full-page capture, RM
E-003	E-003_platformA_listing_northbend.png	cb32b016c86763e4d5997acd0a92f4eda211eee3f76c151e37abaa57e0314703	12 May 10:04 CDT, full-page capture, RM
E-005	E-005_platformB_listing_cedarflats.png	b69c2f5e9e048fd7421780da9b9513b427bc89353faebe51924ebb60d3bdf90c	12 May 11:16 CDT, full-page capture, RM
E-012	E-012_platformA_profile_cedarflats.png	b5421183507f05de7c6e607fd41c8d09f19dd386c0ae99a72deb9db104155629	12 May 12:26 CDT, full-page capture, RM
E-014	E-014_platformB_profile_brightfork.png	c6a607f2f86c90f1740c2d0b86e909001d1f5468e4ef36e1e44daeecbd234534	12 May 12:38 CDT, full-page capture, RM
E-018	E-018_platformB_listing_pinehollow.png	4f1c88a0b7e35d92cc0417ab6e9d2f5081b3a76c4d5e9f02138a7b6c5d4e3f21	12 May 11:31 CDT, full-page capture, RM
E-021	E-021_filestash_page.png	b69d3b364447d0f88f0b568107e055fcc8140248cb647c17e7ac2424341f5229	12 May 13:02 CDT, full-page capture, RM
E-022	E-022_LO-SPEC-4412-RidgelineTent-rev3.pdf	9f2e4b7c1a836d05e4c19f8b2d7a06c35e81f4b9a2c7d0e36f5a8b1c4d9e2f70	12 May 13:11 CDT, original file downloaded, RM
E-031	E-031_registry_cedarflats_trading.pdf	9131a4b820876d2c35b5fe010e9ade267c5ef044af9936d4	12 May 14:22 CDT, official copy downloaded, RM

Item ID	File name	Recorded SHA-256	Recorded acquisition
		bb53b8701bcba078	
E-051	E-051_client_email_ferrera.eml	b4a5b70732f5c23fb12648c5e2e4a0d34f1b8e21c0369acd1ab7cda5cf51205f	13 May 09:05 CDT, provided by client, transferred by encrypted link, RM

8.4.4 The Provided Custody Narrative

The previous investigator also left these notes describing what happened to the evidence. Several described actions are improper. Identify them.

1. 12 May. Captured the listings and profiles over the course of the morning. Hashed everything at the end of the day so the hashes were all recorded together in one sitting.
2. 12 May. Downloaded the specification document. Opened it to confirm it was the right document and that the control number matched, then hashed it.
3. 12 May. Stored all files in a folder on the case drive named "evidence." Made a second copy in a folder named "working" for analysis.
4. 13 May. The client sent the email file by encrypted link. Downloaded it, hashed it, and recorded the hash. Did not ask the client for the hash on their side.
5. 14 May. Annotated E-001 and E-003 with arrows highlighting the identical product image, so that counsel could see the comparison. Saved over the original files.
6. 15 May. A colleague needed to review the listings. Copied the evidence folder to a shared network location so they could look at it, and mentioned it to them in a hallway conversation.
7. 16 May. Sent E-022 to counsel by email attachment. Did not record the transfer because counsel is on the same team.
8. 18 May. Noticed that E-018 was missing from the folder. Assumed it had not copied correctly, so recaptured the listing from the platform and gave the new capture the same item ID.
9. 20 May. Verified hashes for the four items that seemed most important. All matched. Did not verify the rest because nothing appeared wrong.
10. 22 May. Deleted three earlier working copies during a cleanup of the case drive to save space.

Steps

Step 1 (25 min). Hash every file in the provided set using one of the commands in 8.4.2. Record the

file name, full hash, file size, the tool you used, and the date and time on Worksheet 4A.

Step 2 (25 min). Reconcile against the manifest on Worksheet 4B. For every manifest entry and every file present, record: present in folder, present in manifest, hash match, and status. You should find three distinct categories of discrepancy.

Step 3 (30 min). Build the chain of custody record on Worksheet 4C. Create an acquisition entry for every item actually present, using the manifest's acquisition details where they exist and flagging where they do not. Then add every subsequent custody event you can establish from the narrative in 8.4.4, in sequence, marking each entry as either properly documented or reconstructed from the narrative.

Step 4 (30 min). Handling error analysis on Worksheet 4D. Go through the ten narrative items. For each: is the described action proper or improper, which principle it violates, what the consequence is, and what should have been done instead. At least seven of the ten contain a problem.

Step 5 (20 min). Mismatch and discrepancy report on Worksheet 4E. For each discrepancy from Step 2, write the formal report entry: what was expected, what was found, when and by whom it was discovered, the possible explanations, the determination process you would follow, the items and findings affected, and the impact statement you would put in the final report.

Step 6 (20 min). Design the corrected custody arrangement on Worksheet 4F: item ID scheme, hashing procedure and timing, preservation and working copy structure, storage and access control, transfer procedure, verification schedule, production log, and retention and disposal. Address the sole-investigator problem carried from Lab 2.

Step 7 (15 min). Write the custody exception memorandum, half a page, addressed to counsel: what exceptions exist in this evidence set, which items are affected, which findings would be affected, and your recommendation on whether the affected items can be relied upon.

Worksheet 4A: Hash Record

File name	Computed SHA-256 (full)	Size (bytes)	Tool	Date/time computed	By
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Worksheet 4B: Reconciliation

Item ID	In folder?	In manifest?	Hash match?	Status	Discrepancy type
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Worksheet 4C: Chain of Custody Record

Entry #	Item ID	Event type	Date/time	From	To	Purpose	Method	Hash before	Hash after	Result	Location after	Documented or reconstructed	Notes

Worksheet 4D: Handling Error Analysis

Narrative #	Proper or improper	Principle violated	Consequence	What should have happened

Worksheet 4E: Discrepancy Report

Discrepancy	Expected	Found	Discovered when/by	Possible explanations	Determination process	Items affected	Findings affected	Impact statement

Worksheet 4F: Corrected Custody Arrangement

Element	Specification

Expected Outcomes

The three discrepancies. A correct reconciliation finds all three, and they are of different kinds:

1. **A hash mismatch on E-022.** The manifest's recorded hash for the specification PDF does not match the computed hash of the file present. This is the central finding. The correct handling is to report it, not resolve it: record both values, note that the mismatch means either the file changed after the manifest was written or the manifest value is wrong, and observe that narrative item 2 (the file was opened before hashing) plus narrative item 7 (an untracked email transfer) are both candidate explanations. **A student who "corrects" the manifest to match the file has done the one thing this lab exists to prevent.**
2. **A manifest entry with no corresponding file: E-018.** The manifest records an item that is not in the evidence set. Narrative item 8 explains what happened and makes it worse: a replacement capture was given the same item ID, which means the ID no longer identifies a single item and any reference to E-018 is now ambiguous.
3. **A file present with no manifest entry: E-040.** The listing export CSV exists in the folder and appears nowhere in the manifest. It is therefore not evidence, it is a file; nothing establishes where it came from, when, or by whom.

The handling errors. At minimum, students should identify:

#	Problem
1	Hashing at the end of the day rather than at acquisition. The integrity record does not cover the intervening hours
2	Opening the PDF before hashing. Any change caused by opening is inside the unhashed window, and this is a candidate explanation for the E-022 mismatch
3	Not proper by itself, and incomplete: two folders is the right idea, but nothing establishes that the preservation copy is never opened, no hashes are recorded for the working copies, and there is no access control
4	Acceptable but weak: without the client's hash there is no verification of the transfer, so the record begins at receipt rather than at the client's system
5	Annotating and saving over the originals. This destroys the originals and changes their hashes. It is the most serious error in the narrative after the E-018 ID reuse
6	Copying evidence to a shared location with no access control and no record, and communicating it verbally. Unlogged access by an unknown number of people
7	An untracked transfer. "Same team" is not an exception; the transfer period is unaccounted for and the copy sent to counsel is untracked
8	Reusing an item ID for a different item. The single worst error: two different files now share one identifier, and the recapture is a different item collected on a different date showing a possibly different page state
9	Selective verification. Verifying only the important-looking items means the others' continuity is unestablished, and successful verifications on the rest are exactly the evidence of continuity that is now missing
10	Unauthorized destruction during cleanup, with no record of what was deleted

The exception memorandum should conclude approximately: E-022's integrity cannot be established

between acquisition and the present, and the item should not be relied upon for any finding without either a verified alternative copy or an acknowledgment of the gap; E-018 does not exist as a coherent item and any reference to it must be withdrawn or re-identified; E-040 is unlogged and must be either properly acquired again with documentation or excluded; E-001 and E-003 have been altered and the originals are lost, which is disclosable; and the overall custody record contains gaps that must be disclosed rather than reconstructed.

Discussion Questions

1. You found a hash mismatch. Describe the two paths available to you at that moment and why one of them ends a career.
2. Narrative item 8 reuses an item ID. Explain, to a non-specialist, why that is worse than simply losing the file.
3. Narrative item 9 verified only the important items. Explain why the successful verifications you did not perform are the loss, not the failed ones.
4. Which of the ten narrative actions would you expect to be most common in real practice, and what would prevent it?
5. Your exception memorandum tells counsel that some evidence cannot be relied upon. Explain why delivering that memorandum makes you more valuable rather than less.

Grading Rubric: Lab 4 (80 points)

Criterion	Weight	4 points looks like
Hashing performed on all files	x2	All ten, full 64-character hashes, tool and time recorded
Full hashes recorded, not truncated	x2	No truncation anywhere
E-022 mismatch detected	x4	Found, and both values recorded
E-022 mismatch reported, not resolved	x4	Manifest not "corrected"; explanations and process stated
E-018 missing item detected	x3	Found, and the ID reuse consequence identified
E-040 unlogged item detected	x3	Found, and correctly characterized as not evidence
Custody record construction	x3	Acquisition entries plus narrative events, sequenced, documented versus reconstructed marked
Handling error analysis	x4	Seven-plus errors with principle, consequence, and remedy
Item 5 (annotated originals) identified	x2	Recognized as destroying originals and changing hashes

Criterion	Weight	4 points looks like
Item 8 (ID reuse) identified as most serious	x3	Reasoning about identifier ambiguity
Item 9 (selective verification) reasoning	x2	Recognizes that missing successful verifications are the loss
Discrepancy report quality	x3	All fields, including impact on findings
Corrected custody arrangement	x3	Complete, with the sole-investigator weakness addressed
Exception memorandum	x4	Honest, item-specific, with a reliance recommendation

Total possible: 168 weighted. Passing: 116.

Lab 5: Case Management From Initiation to Closure

Time required: 3 to 3.5 hours **Format:** Individual **Difficulty:** Advanced

Objectives

1. Operate a case file structure with task and lead registers.
2. Manage a case through review points, reprioritization, and a scope decision.
3. Reconcile evidence and custody at closure.
4. Produce a closure memorandum and a complete case package.

Prerequisites

- Section 1.8 and comparison tables 2.7 and 2.8.
- Labs 1 through 4 complete. Bring every artifact.

The Six-Week Event Feed

These events occurred during the investigation. Manage them.

Week	Event
1	Plan approved at version 1.0. Collection begins. 46 listings captured across two platforms; 11 seller accounts identified.
1	County records request for the Unit 6 warehouse district address sent (three-week latency).

Week	Event
1	Client provides the mark registration numbers but says the authorized dealer list will take a week.
2	Two of the eleven seller accounts are matched to registered entities. Four are matched to declared business names with no registration located. Five have no declared business name.
2	A listing captured in week one is deleted. Preservation copy retained.
2	Whitfield asks informally whether you could also look at whether any authorized dealer is reselling into the grey market.
3	Review point 1. Hash verification run: the E-022 mismatch from Lab 4 is discovered.
3	The authorized dealer list arrives, one week late. Cross-check shows none of the eleven sellers is an authorized dealer.
3	Two of the eleven seller accounts are found to share a declared business name with a third-party fulfilment company, which also serves several unrelated legitimate sellers.
4	County records arrive. The Unit 6 address is a multi-tenant warehouse with 22 units under one owner.
4	The file-sharing page hosting the specification document is removed. Preservation copies retained.
4	A new listing appears on a third platform, which was not in the original in-scope platform list.
5	Review point 2. Objective 3 (document history) is blocked: the earliest public appearance cannot be established from available sources beyond a "41 days ago" relative timestamp on a now-removed page.
5	The client asks for a preliminary briefing before the formal report.
5	Analysis identifies that four listings share an identical product image, and that the four accounts were created within an 11-day window.
6	Report drafted, reviewed, and delivered. Briefing held.
6	Whitfield asks two follow-up questions and

Week	Event
	requests one change to the report's wording.
6	Case closure.

Steps

Step 1 (20 min). Build the case file structure on Worksheet 5A. Use or adapt the structure in Section 1.8.3. Then map every artifact you have produced across Labs 1 through 4 into it, and identify any artifact that does not yet exist.

Step 2 (30 min). Task register on Worksheet 5B. Populate from Lab 1's task list, then update every row against the event feed: status, completion, blocked, or descoped. Minimum eighteen rows, each citing an objective.

Step 3 (35 min). Lead register on Worksheet 5C. Every lead arising from the event feed, minimum twelve. For each: lead ID, source item or event, description, in scope or not, priority, status, and **a recorded disposition for every closed lead**. The week 4 third-platform listing, the week 2 grey-market request, and the week 3 fulfilment company are all leads requiring dispositions of different kinds.

Step 4 (25 min). Scope decisions on Worksheet 5D. Two events require formal handling: Whitfield's week 2 grey-market request and the week 4 third-platform listing. For each, apply the scope change procedure: what you record, what you do not do, what an amendment would say, what it would cost and displace, and your recommendation. Then write the two sentences you would send Whitfield for each.

Step 5 (25 min). Review point records on Worksheet 5E. Document both review points: progress against each objective, evidence integrity status, documentation reproducibility check, scope check, and any reprioritization with its trigger. Review point 1 must record the E-022 discovery and the resulting actions.

Step 6 (25 min). Closure reconciliation on Worksheet 5F. Work the closure checklist from Section 1.8.7: every objective assessed with a basis, every lead dispositioned, the evidence inventory reconciled against the custody record, custody exceptions listed, the documentation reproducibility test result, delivery and receipt, feedback recorded, and retention applied with a disposal date.

Step 7 (30 min). Closure memorandum. One page. What was accomplished, what was not and why, what remains open, what exceptions exist, what was descoped and by whose decision, the retention and disposal arrangement, and a short section on what a future investigator picking this up would need to know first.

Step 8 (15 min). Assemble and index the case package. Produce a one-page index listing every artifact, its location in the structure, its version, and its date. Then state which three artifacts a reviewer would open first and why.

Worksheet 5A: Case File Map

Folder	Artifact	Source lab	Version	Date	Present?
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Worksheet 5B: Task Register

Task ID	Obj ID	Description	Owner	Due	Status	Output	Notes
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Worksheet 5C: Lead Register

Lead ID	Source event/item	Description	In scope?	Priority	Status	Disposition and basis
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Worksheet 5D: Scope Decisions

Event	Recorded as	Action taken	Amendment text	Cost and displacement	Recommendation	Consumer message
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Worksheet 5E: Review Records

Review #	Date	Objective status	Evidence integrity	Documentation check	Scope check	Reprioritization and trigger	Actions
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Worksheet 5F: Closure Reconciliation

Closure criterion	Status	Basis	Exceptions
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Expected Outcomes

- Every Lab 1 through 4 artifact mapped into the structure, with the missing ones identified. Most students discover that they have no communications record and no production log.
- **The week 2 grey-market request is the primary scope test.** It is out of scope (authorized dealers are excluded by the Lab 1 scope statement), it came informally, and the correct handling is: record the request, do not collect, prepare an amendment describing what it would add and what it would displace, and put the decision back to Whitfield in writing. Students who simply start looking at dealers have failed the criterion; students who refuse without offering the amendment path have failed differently.
- **The week 4 third-platform listing is the second scope test**, and it resolves differently: it is within the activity and subject scope but outside the stated platform list, so it is a modest

amendment rather than a refusal, and the recommendation should be to amend, with the displacement named.

- **The week 3 fulfilment company is a discrimination test carried from Chapter 5.** Two accounts sharing a fulfilment provider that also serves unrelated legitimate sellers is a non-discriminating shared attribute. The lead should be dispositioned as closed with that reasoning recorded, not drawn as a relationship.
- **The week 4 county records finding is a similar trap.** A multi-tenant warehouse with 22 units under one owner means a shared address is weakly discriminating at best, and any relationship inference from the shared address must be qualified accordingly.
- **Objective 3 is blocked, not failed.** The closure memorandum must say so, state what would resolve it (the file-sharing platform's records, obtainable only by legal process the engagement does not authorize), and not quietly report it as complete.
- Every lead has a disposition. A lead register with open leads at closure is acceptable only if each is explicitly recorded as open and handed off.
- The closure memorandum must disclose the E-022 custody exception. A memorandum that omits it fails the criterion regardless of how good the rest is.
- The three artifacts a reviewer opens first should be the plan, the closure memorandum, and the evidence inventory with its custody record. Students who name the report are thinking like an author rather than a reviewer.

Discussion Questions

1. Whitfield asked informally for something out of scope. Write the exact sentence you would send. What makes an informal request harder to handle than a formal one?
2. Objective 3 is blocked by an authorization limit rather than by a lack of effort. How do you report that without it sounding like an excuse?
3. Two accounts share a fulfilment company. Explain why that is not a relationship, using the Chapter 5 test.
4. Your closure memorandum discloses a custody exception. Argue that this increases the value of the package.
5. A future investigator picks up this case in a year. What is the first thing they need to know, and is it written down?

Grading Rubric: Lab 5 (84 points)

Criterion	Weight	4 points looks like
Case file structure and mapping	x2	Complete; missing artifacts identified
Task register maintenance	x3	Eighteen-plus rows, updated

Criterion	Weight	4 points looks like
		against every event, objectives cited
Lead register completeness	x3	Twelve-plus leads, every closed lead dispositioned with a basis
Grey-market scope decision	x4	Not collected; amendment prepared; decision returned to the consumer in writing
Third-platform scope decision	x3	Correctly resolved as an amendment with displacement named
Fulfilment company discrimination	x3	Closed as non-discriminating, with reasoning
Multi-tenant address handling	x2	Shared address qualified appropriately
Review point records	x3	Both documented across all five dimensions
E-022 handled at review point 1	x3	Discovery, actions, and impact recorded
Closure reconciliation	x3	Every criterion assessed with a basis
Objective 3 reported as blocked	x3	With the resolving route named
Closure memorandum honesty	x4	Accomplishments, failures, exceptions, descoping, handoff
Case package index	x2	Complete; correct three artifacts named for a reviewer

Total possible: 152 weighted. Passing: 105.

Part 9: Capstone Assignment

The Cordelia Falls Water Authority Investigation

Weight: 35% of chapter grade **Time required:** 6 to 9 hours **Format:** Individual **Deliverables:** A complete investigation package, specified in Section 9.4

9.1 The Scenario

Everything below is fictional.

You have been engaged by the **Cordelia Falls Water Authority**, a public body serving 96,000 residents, governed by a five-member elected board. Your engagement letter is signed by the board chair and the Authority's general counsel.

The trigger. Three matters arrived within one month and the board has asked whether they are related.

Matter A: the meter procurement. The Authority awarded a \$4.6M smart meter replacement contract 14 months ago to **Alder Point Metering LLC**. The rollout is 40% behind schedule. Twelve change orders totalling \$780,000 have been approved by the operations director. An anonymous letter to the board alleges that the vendor was "pre-selected."

Matter B: the data exposure. A resident contacted the Authority to report that a spreadsheet containing customer account numbers, service addresses, and meter readings for approximately 4,000 accounts was available on a public file-sharing site. The Authority's IT contractor confirmed the file appears to originate from a report the Authority generates monthly and provides to Alder Point Metering under the contract.

Matter C: the billing complaints. Over eight weeks, 61 residents have complained of billing errors following meter replacement. The operations director has attributed these to "expected transition issues."

The decision. The board must decide at its meeting in **seven weeks** among:

1. Continue the contract with corrective action on process.
2. Suspend the contract pending a formal audit.
3. Terminate for cause and re-tender.
4. Refer one or more matters to the state auditor, the attorney general, or law enforcement.

Complications.

- The Authority is a public body subject to open records law. Your report will very likely become public.
 - The 4,000 affected residents are private individuals whose data is in the exposed file.
 - The operations director, **T. Adeyemi**, is a 16-year employee who has stated that the change orders were operationally necessary and that the pre-selection allegation is baseless.
 - A board member has publicly called the situation "a cover-up."
 - Notification obligations may apply to the data exposure, with a statutory clock that may already be running.
 - Your engagement authorizes open source and public records investigation, examination of Authority records provided to you, and interviews of Authority staff arranged through counsel. It does not authorize examination of staff personal accounts or devices, contact with Alder Point Metering, or any access to the file-sharing site's non-public information.
-

9.2 Available Material

You will not be given an evidence packet. **Designing the collection is the assignment.** You are told what exists and is obtainable:

Available	Notes
The procurement file: solicitation, bids, evaluation scores, award, contract	Authority records, provided by counsel
All twelve change orders with supporting documentation	Authority records
Board minutes and agendas, 24 months	Public records
The Authority's procurement policy and delegation thresholds	Public records
The Authority's published budgets and audited financial statements	Public records
The monthly report generated for Alder Point, current and prior versions	Authority records
The Authority's IT contractor's incident notes	Authority records, third-party generated
The file-sharing page (still live at initiation)	Public
The exposed file itself	Public, and it contains 4,000 residents' personal data
The 61 billing complaints	Authority records, containing residents' personal data
Corporate registry, court, UCC, property, and licensing records for Alder Point and any affiliate	Public
Alder Point's website, current and archived	Public
Trade press covering smart meter procurements and typical schedule performance	Public
Comparable authorities' published contract amendments and rollout schedules	Public
Staff interviews, arranged through counsel	Authorized
The anonymous letter	Authority record

9.3 Guidance on the Substance

Three matters, three different evidentiary states, and possibly three different answers. A weak package treats them as one story. A strong package plans them as three lines of inquiry with shared administration, and its scope statement says so.

The exposed file is the hardest planning problem in this assignment. You need it to establish what was

exposed, and it contains 4,000 residents' personal data, and your report will become public. How you plan its collection, handling, storage, description, and disposal is the single most heavily weighted element of the capstone.

The comparative question is available for Matters A and C and is likely decisive for both. The negative question is available for all three.

9.4 What You Must Produce

A complete investigation package, in nine parts.

Deliverable 1: Investigation Plan (20 points)

A full plan, six to twelve pages, containing:

- Authority and initiation, including what the engagement does and does not permit, and the case number.
- One strategic goal tied to the board's seven-week decision.
- A scope statement bounding all four dimensions, with **minimum eight exclusions each with a reason**, minimum six method boundaries, and minimum five assumptions. The exclusions must address staff personal accounts, residents whose data is in the exposed file, the complainants, and the anonymous letter's author.
- **Minimum six objectives**, each measurable, each with a rationale, success criterion, completion test, dependencies, owner, and deadline, and each completable by a negative finding.
- A task register of minimum twenty-four tasks, each citing an objective.
- A priority matrix using a stated method with a perishability override, a latency overlay, and a named drop order of at least three items.
- A scope change procedure with a default of stop.
- Risks and constraints, minimum six, with mitigations, including the notification clock.
- Closure criteria, approval block, and revision history at version 1.0.

Deliverable 2: Evidence Collection Plan (16 points)

- Minimum twenty-four rows, minimum seven Chapter 3 source categories, minimum six rows from official and compelled sources.
- Every row with: evidence sought, category, source, collection method appropriate to the type, perishability, legal and authorization basis with the four tests applied, documentation required at collection as an actual field list, preservation procedure, and expected coverage boundary written in advance.
- At least three rows blocked pending authorization, identified as such.

- **A dedicated handling procedure for the exposed file**, covering: whether and how it is collected, how it is stored, who may access it, how it is described in the report, whether any of its content is reproduced anywhere, how the resident data is protected from disclosure through your own product, and its disposal.
- Handling procedures for at least six anticipated situations, written in advance.

Deliverable 3: Documentation Plan and Templates (12 points)

- The documentation set for this investigation: which artifacts, formats, naming convention, storage location, and review cadence.
- Completed template headers for: investigator notes, search log, capture record, evidence log, decision log, lead register, communications record, and production log.
- A written documentation standard covering contemporaneity, attribution, negatives, correction handling, observation and conclusion separation, and the evidence versus work product distinction.
- The reproducibility test procedure and when it will be run.

Deliverable 4: Chain of Custody Procedures (14 points)

- The complete custody procedure: item ID scheme, hashing algorithm and timing, preservation and working copy structure, storage and access control, transfer procedure with both-ends hashing, verification schedule, production log, and retention and disposal.
- **Special provisions for the exposed file** as a custody problem: it is evidence, it must be preserved, and it is a disclosure risk in your own possession.
- A custody exception procedure specifying what happens when a break is found, who is notified, and how impact is assessed and reported.
- A worked example: a completed custody record for five hypothetical items showing acquisition, preservation copy creation, one transfer to counsel, one scheduled verification, and one production event.
- Identification of the control weakness created by your role structure, with a mitigation.

Deliverable 5: Case Management Design (12 points)

- The case file structure, populated with the artifacts this investigation will produce.
- Task register and lead register formats with initial content.
- Roles and, where you are performing several, the resulting control weaknesses.
- Review points: when, covering what, producing what record.
- The communications plan: who is the single point of contact, what cadence, what is written.
- Closure criteria and the closure memorandum outline.

- A note on whether you would recommend manual or software-based management for this case, with reasoning.

Deliverable 6: Simulated Execution Record (10 points)

You will not actually run the investigation. Produce a simulated but internally consistent record of **one week** of execution, covering minimum twelve activities, including at least: two searches with one negative result, three captures, one download of an original file, one hash verification, one out-of-scope encounter, one visible correction, and one decision record. Documentation must meet the standard from Deliverable 3.

Deliverable 7: Investigative Report (16 points)

Maximum four pages plus annexes. Addressed to the board through counsel, written in the knowledge that it will become public.

Required: a bottom line of no more than five sentences carrying a judgment, a confidence level, and which course of action the evidence supports; **the three matters treated separately with their own confidence levels**; facts separated from assessments with every fact cited to an item ID; the relationship findings worded without asserting intent; the comparative findings for Matters A and C; assumptions; ranked gaps each with a resolving source and a note on whether your engagement permits reaching it; minimum eight recommendations that are specific, assignable, and time-bound, including at least two executable this week, at least two that are referrals or information requests, and at least one addressing the notification obligation; what would change this assessment; and annexes comprising the evidence inventory, custody record, and ethics annex.

Because this is a planning chapter, the report is assessed on structure, sourcing discipline, characterization, and honesty about limits rather than on analytic depth. State your findings as they would follow from the collection you planned, and mark any finding you are asserting for the purposes of the exercise as such.

Deliverable 8: Closure Package (8 points)

- A closure reconciliation against your stated criteria.
- A closure memorandum: accomplishments, shortfalls with reasons, open items, exceptions, descoping decisions and who made them, retention arrangement with a disposal date, and a handoff note.
- A case package index listing every artifact, location, version, and date.
- **A disposal plan for the exposed file** specifying when and how it is destroyed, who authorizes it, and how the destruction is recorded.

Deliverable 9: Decision Justification Log (2 points)

Minimum eighteen rows: each significant planning decision, the alternatives, why you chose as you did, and what would have changed it. Absence caps the capstone at 70.

9.5 Capstone Grading Rubric (110 points, scored to 100)

Deliverable	Criterion	Points
1. Plan	Authority, engagement limits, strategic goal	3
	Four scope dimensions bounded	3
	Eight-plus exclusions with reasons, covering all four required subjects	4
	Method boundaries and assumptions	2
	Six-plus measurable objectives with success criteria and completion tests	4
	Negative-path completability on every objective	2
	Task register, twenty-four-plus, objectives cited	2
	Priority matrix with perishability override and latency overlay	3
	Drop order, three-plus named items	2
	Scope change procedure with default stop	2
	Risks including the notification clock	2
2. Evidence plan	Row volume and category breadth	3
	Collection method appropriateness per type	4
	Four tests and authorization status per row	3
	Documentation required at collection, as field lists	2
	Coverage boundaries in advance	2
	Exposed-file handling procedure	5
	Anticipated-situation procedures, six-plus	2
3. Documentation	Artifact set, formats, naming, storage, cadence	4

Deliverable	Criterion	Points
	Template headers, eight artifacts	4
	Written documentation standard	3
	Reproducibility test procedure	2
4. Custody	Complete procedure, all elements	5
	Exposed-file custody provisions	4
	Exception procedure	2
	Worked five-item custody record	3
	Role-structure weakness identified with mitigation	2
5. Case management	Case file structure populated	3
	Task and lead register formats with content	3
	Review points and communications plan	3
	Closure criteria and manual-versus-software recommendation	3
6. Execution record	Twelve-plus activities, internally consistent	4
	All required element types present	3
	Documentation standard met, including the visible correction	3
7. Report	Bottom line: five sentences, judgment, confidence, course of action	3
	Three matters treated separately with distinct confidence	4
	Facts cited and separated from assessments	3
	Comparative findings for A and C	2
	Gaps ranked with resolving sources and authorization notes	2
	Recommendations, including notification	2
8. Closure	Reconciliation and memorandum honesty	4

Deliverable	Criterion	Points
	Package index	1
	Exposed-file disposal plan	3
9. Decision log	Eighteen-plus rows, substantive	2
Total		110, scored to 100

Grade bands: 90 to 100 exemplary, 80 to 89 proficient, 70 to 79 competent, 60 to 69 developing, below 60 not yet passing.

Automatic deductions:

- Reproducing, quoting, or including any resident's account number, address, or personal data anywhere in any deliverable, including as an example: **minus 30**. The exposed file is evidence and must be described, counted, and characterized, never reproduced. This is the assignment's central test.
- Failing to address the exposed file as both evidence and a disclosure risk in your own possession: minus 15.
- No exposed-file disposal plan: minus 10.
- Treating the three matters as one undifferentiated finding: minus 10.
- Any judgment without a confidence level: minus 5 per instance.
- Asserting that any individual acted with improper intent: minus 10 per instance.
- Planning collection of staff personal accounts or contact with the vendor, which the engagement excludes: minus 10 per instance.
- No coverage boundary on a negative finding: minus 3 per instance.
- Objectives phrased so they can be completed only by a positive finding: minus 4 per instance.
- Missing decision justification log: capped at 70.

Guidance on the expected shape of a strong package. The three matters plan differently. Matter A is records-heavy, stable, and comparative: the procurement file, the change order distribution against delegation thresholds, board minutes, and peer authorities' schedule and amendment performance will carry it, and the anonymous letter's allegation is testable against the evaluation scoring rather than against anyone's character. Matter B is the most perishable and the most legally exposed: the file-sharing page must be captured immediately, the file itself is both essential evidence and a liability in your custody, the notification clock may already be running, and the correct handling is to establish what was exposed and when without ever reproducing content. Matter C is comparative first: 61 complaints over eight weeks means nothing until you know the complaint rate during comparable transitions and the total number of meters replaced. And the relationship question the board actually asked, whether the three matters are related, is answerable only after all three are separately

established, which the plan should say explicitly rather than promising a unified narrative.

Part 10: Answer Keys

Instructors may wish to remove this section from student copies.

10.1 Short Answer Key

1. An investigation plan is a written document produced before substantive work stating what the investigation covers and excludes, what it must accomplish, in what order, using what evidence, documented how, managed by whom, and finished when. Sections include authority and initiation, background, scope, objectives, priorities, evidence collection plan, documentation plan, custody procedures, case management, legal and ethical framework, risks, closure criteria, and approvals with revision history.
2. Without scope, the investigation expands without limit. Without objectives, there is effort without completion. Without priorities, effort is spent on the wrong things by the deadline. Without documentation and custody, correct findings cannot be defended. Above all of them: without a plan there is nothing to review, so a reviewer cannot assess whether the work was competent.
3. Proportional planning means the plan's length scales with the investigation but every section still appears. A two-day investigation gets a one-page plan with compressed versions of every section. Deleting a section does not save time; it moves the cost downstream, and most small investigations that fail do so because nobody spent twenty minutes on the plan.
4. Subject (which entities, systems, or activities), temporal (what period), geographic or jurisdictional (which places, jurisdictions, or platforms), and activity (what types of conduct are examined). Examples should be specific enough to be testable.
5. Exclusions prevent drift, protect the parties who are out of scope, tell the consumer what they are not buying, and tell a later reviewer that an omission was a decision rather than an oversight. They are the most omitted part of a scope statement and the part that does the most work.
6. Scope creep is gradual expansion beyond the stated boundary without a decision to expand. Causes include an interesting thread appearing, a finding suggesting a wider problem, informal additions from the consumer, an adjacent question being cheap, and the desire for a complete picture. The control is a written scope change procedure whose default is stop.
7. Because an investigator who works an out-of-scope thread first and seeks approval afterward has already created the exposure the procedure exists to prevent. The collection has occurred, it may be unauthorized, and the material's quality does not cure the defect.
8. An objective is measurable when it names a deliverable, standard, or determination that can be checked, such that two people looking at the work would agree on whether it is done. "Understand the problem" is unmeasurable; "produce an inventory of all in-scope listings, captured, hashed, and logged, as of the capture date" is measurable.

9. Because an objective phrased "establish that X" can only be completed by finding X, which means the investigator is being paid to reach a particular result. "Determine whether X can be established from available sources, recording either the finding or a documented negative with its coverage boundary" can be completed either way and removes that pressure.

10. Any four dimensions from: number (one versus dozens), duration (whole investigation versus hours), owner (consumer and lead versus individual investigator), rate of change (rarely by amendment versus constantly), measurement (whether the decision was supported versus whether the output exists), and failure signature (investigation succeeds but the decision is unsupported versus a task complete that served no objective).

11. A success criterion states what completion looks like; a completion test states how someone else verifies it. The second matters because without it "done" is a claim by the person who did the work, whereas with it done is a checkable state, and in organizations that review investigations the completion test is what a reviewer actually performs.

12. Any four of decision impact, perishability, dependency, cost and latency, risk, and external deadline. Decision impact is the default because the purpose of the investigation is to support a decision, so the ranking that maximizes decision support is the correct primary basis.

13. The perishability override is the rule that perishable evidence is captured first regardless of its impact ranking, because it disappears on its own schedule. Examples: marketplace and social listings, days to weeks; platform profiles, days; system and network logs, days to months by retention policy; CCTV, days to weeks; volatile system state, minutes.

14. Because if the drop order is not written down, the deadline decides, and the answer in practice becomes whatever happened to be last on the list. Naming the descoped items in advance converts a resource shortfall from a crisis into a plan, and lets the consumer object while there is still time.

15. Collection method, documentation required at the moment of collection, legal and authorization basis, and preservation procedure. Without them a collection plan produces material; with them it produces evidence.

16. Because printing produces a different file: different metadata, different internal structure, and a different hash from the file that exists on the source system. The printed copy may still be useful, but it is no longer the original, and any metadata evidence the original carried is destroyed.

17. Integrity mechanism: physical uses seals, packaging, and secure storage, digital uses cryptographic hashes. Alteration visibility: physical alteration is usually visible, digital alteration is usually invisible. Working practice: with physical evidence you minimize handling of the original, with digital evidence you never work on the original and always use a verified copy.

18. A matching hash establishes that a file is bit-for-bit identical to the file that was hashed at acquisition. It does not establish that the content is truthful, that the source was authentic, that the collection was authorized, or that the file was not already altered before you collected it.

19. Rule 7 is that if it is not written down, it did not happen. Substantive reasons: reproducibility, since work that cannot be reproduced cannot be verified and an unverifiable finding is an opinion; memory,

which is unreliable and degrades while feeling reliable; and challenge, which is normal and targets process, so only documentation answers it.

20. Chain of custody exists to answer whether an item is the same item, in the same state, as when it was collected. It breaks through gaps in the record, unlogged access, hash mismatches, late or absent acquisition hashing, shared credentials, untracked copies, working on the original, retroactive entries, and anonymous entries. The correct response is to stop using the item, record the discovery, not reconstruct the missing entries, determine the extent, notify, assess the impact on each affected finding, and report the break in the final report.

10.2 Multiple Choice Key

#	Answer	Note
1	b	A written document produced before substantive work
2	b	Exclusions
3	c	Unbounded on the temporal and geographic edges
4	b	Stop, record the observation, seek a decision
5	b	Two people could not agree it is complete
6	b	Completable only by a positive finding
7	c	Each task cites the objective it serves
8	c	Decision impact
9	b	Perishable capture first; slow item initiated immediately
10	b	A different file with different metadata and hash
11	c	Bit-for-bit identity with the acquisition-time file
12	b	Creates a new file, changes the hash, destroys the original
13	b	Report, determine extent, assess impact, state it
14	b	Documentation in small, coordination in large
15	b	Objectives assessed, leads dispositioned, inventory

#	Answer	Note
		reconciled, retention applied

10.3 True or False Key

#	Answer	If false, why
1	False	It needs a one-page plan; sections compress but none is deleted
2	True	
3	False	Priorities are set in the plan, because during execution the loudest thread wins
4	True	
5	False	A late hash proves nothing about the period before it was computed
6	False	The preservation copy is never opened; all analysis uses working copies
7	False	Successful verifications are the evidence of continuity and must be recorded
8	False	Negative results with coverage boundaries are findings, and their absence lets a limitation be read as a conclusion
9	False	A lead that stops being mentioned is lost, not closed; closure requires a recorded disposition
10	True	

10.4 Critical Thinking Guidance

7.1 The most important exclusion is normally patients: the concern is about staff conduct and a scheduling system, and patient records are neither necessary nor proportionate to it. Strong answers also exclude staff personal accounts and devices, the complainant's identity, and any minor, and set method boundaries prohibiting access to the scheduling system beyond what compliance authorizes, contact with any patient, and any collection outside the stated period.

7.2 Record the observation in the decision log describing what was seen and where, with no substantive collection. Prepare an amendment stating what the permit review would add, the objectives it would create, the estimated effort, and what it would displace from the current priority list. Return the decision to the consumer in writing. Do not begin the permit research first.

7.3 Defects: subject unbounded ("all activity" and "associates"), temporal boundary vague ("recent years"), no geographic or jurisdictional boundary, no activity boundary, no exclusions, no method boundary ("all available sources" invites legal exposure), and the closing clause is an objective that presupposes an actor. The rewrite should bound all four dimensions, add exclusions and method boundaries, and restate the purpose as a determination rather than an attribution.

7.4 (a) "For each account in the inventory, determine whether an operator can be identified from public records, recording either the identification with supporting records or a documented negative with its coverage boundary." (b) "Determine what the vendor's public filings, liens, judgments, and licensing status disclose about its financial position, and produce a written summary with the sources and the limits of what public records can establish." (c) "Determine the earliest identifiable public appearance of the document and produce a sourced timeline of subsequent appearances, or a documented statement of why the earliest appearance cannot be established." (d) "For each of the five claims in the company's published statements, determine whether public records support, contradict, or are silent on the claim."

7.5 The strategic goal must name a decision and a date. At closure without objectives, the case would have a list of completed tasks and no way to state what was accomplished, because nothing defined accomplishment; the consumer's question "did you answer my question" would be unanswerable from the file.

7.6 Two defects: it is unmeasurable ("extent" is undefined and no output is named), and it presupposes that fraud occurred, so it can be completed only by finding fraud. Rewrite: "Determine, for each transaction above the threshold in the period, whether supporting documentation exists and is internally consistent, and produce a schedule of supported, unsupported, and inconsistent transactions with the basis for each classification."

7.7 The four two-week-retention tasks come first regardless of impact. The two records requests are initiated on day one because of latency, though they are not worked first. The three high-impact stable tasks occupy weeks one and two after the perishable captures. The two low-impact interesting tasks are the drop order, named explicitly. The defence should distinguish clearly between what is sequenced first and what is ranked highest.

7.8 Reprioritize, record the trigger, the old rank, the new rank, and the displacement. This is a reprioritization rather than an amendment, because the scope and objectives have not changed; only the order has. If working the task required going outside the scope, it would instead be an amendment.

7.9 The response should explain that an unranked list means the deadline will choose, ask which single objective the consumer would keep if only one could be completed, and then build the ranking from that answer outward. Accepting "all equally important" is a planning failure because it transfers the ranking decision from the consumer, who owns it, to circumstance.

7.10 Rows must specify method appropriate to type: full-page captures plus independent archive for

listings, original file downloads for documents, native exports with recorded queries for registry results. Perishable rows are the listings and profiles. The slow item is typically a county records request or a certified copy. Documentation requirements must be field lists, not "document it."

7.11 Accept the client's screenshots as a client-provided item with its own item ID, recorded provenance, and the limitation that their collection method, timing, and tooling are unknown and unverifiable. Do not treat them as a substitute for your own captures, because you cannot attest to how they were made, whether they are complete, or whether they have been altered. Record the receipt, hash them, and make your own captures of anything still live.

7.12 Method: photograph or capture the terminal screen if permitted, and request an official or certified copy from the custodian in parallel. Documentation: the custodian, the record identifier, the terminal location, the retrieval time, who was present, and the request made. Integrity: hash your own capture immediately at creation, note that no source-side hash is possible, and treat the certified copy when it arrives as the authoritative item with your capture as a contemporaneous record of what was displayed.

7.13 A complete answer names the artifact set, the file formats, a naming convention using item IDs and dates, a single case store with a defined folder structure separating preservation from working copies and evidence from work product, and a review cadence including a mid-investigation reproducibility test.

7.14 Honest reporting is the grading basis. The two artifacts whose absence most commonly causes difficulty are the search log (so nobody can reproduce how something was found) and capture records (so a screenshot has no URL, time, or tool).

7.15 The rewrite should produce multiple entries, each timed, attributed, and specific: which platform, which exact terms, which filters, what result count, which items captured, which hashes recorded, which searches returned nothing and with what coverage boundary. Omitted fields in the original include date, time, time zone, initials, system, terms, filters, counts, item IDs, negatives, and any separation of observation from conclusion.

7.16 Item IDs assigned at acquisition in a single sequence, never reused. SHA-256 computed at acquisition before any other action, recorded with the file size and tool. A preservation copy per item, verified and never opened, in a separate folder; working copies made from it. Access limited and logged. Transfers hashed at both ends with both parties recording. Verification on a stated cadence with all results recorded. The client-provided email requires its own provenance entry recording that it originated outside your control.

7.17 Entries: a verification entry recording the expected hash, the computed hash, the file, the date and time, and the result as a mismatch; and a new entry recording the notification. Do not overwrite the original recorded hash. Notify the lead, the client, and counsel per the plan. Determination: check for transcription error, algorithm mismatch, and whether the file was opened, re-saved, converted, or transferred untracked. The report sentence should state that the item's integrity cannot be established between acquisition and discovery, identify the findings affected, and state whether the item is relied upon.

7.18 Re-collectability is not integrity: a re-collected item is a different item, collected on a different

date, showing a possibly different state, and it does not establish what the page showed when it mattered. Alteration is invisible in digital material, so without a hash there is nothing to detect it. A challenge would target exactly this: not whether the material exists, but whether the specific item relied upon is what the investigator says it is, and without hashes and a single tracked copy that question cannot be answered.

7.19 The two most important structural decisions are separating preservation from working copies at the folder level, so that opening the wrong one requires a deliberate act, and separating evidence from work product, so the evidence inventory remains reviewable. With two investigators, a third decision matters: naming one of them as the custodian so that custody entries have a single owner.

7.20 First: establish authority and write a plan, even retrospectively, marked as such. Then reconcile the 40 items against whatever record exists, hash everything now and record that the hashes date from now rather than acquisition, and identify which items have no provenance at all. What may be unrecoverable: acquisition-time integrity for every item, the search records showing how things were found, and any negative results that were never written down. Tell the consumer plainly which findings are affected and that some items may not be reliable.

7.21 The memorandum should state the three met objectives with their bases, the two unmet with the reason for each (blocked, descoped, or incomplete, distinguished), the two open leads with handoff notes, the custody exception with its impact, and the unresolved records request with what it would have resolved. Honesty is more credible because a reader who finds an omission later discounts everything else, whereas a disclosed shortfall lets the reader calibrate the rest of the package.

Appendix A: Reusable Templates

A.1 Investigation Plan Skeleton

CASE NUMBER: _____ VERSION: _____ DATE: _____
PREPARED BY: _____ APPROVED BY: _____ DATE: _____

1. AUTHORITY AND INITIATION
Authorized by · basis · date · what the engagement permits
and does not permit
2. BACKGROUND
What prompted this, as facts and as a concern
3. STRATEGIC GOAL
One sentence, tied to a named decision and date
4. SCOPE
Subject · temporal · geographic/jurisdictional · activity
Inclusions · Exclusions (with reasons) · Method boundaries ·
Assumptions
5. OBJECTIVES
Each with rationale, success criterion, completion test,
dependencies, owner, deadline
6. PRIORITIES
Method · composite ranking · schedule · drop order
7. EVIDENCE COLLECTION PLAN (Appendix A.4)
8. DOCUMENTATION PLAN (Appendix A.5)

- 9. CHAIN OF CUSTODY PROCEDURES (Appendix A.6)
- 10. CASE MANAGEMENT
 - Roles · task and lead registers · review points · communications · closure criteria
- 11. LEGAL AND ETHICAL FRAMEWORK
 - Four tests · prohibited activities · individuals policy · authorization requirements · retention
- 12. RISKS AND CONSTRAINTS
- 13. SCOPE CHANGE PROCEDURE (default: STOP)
- 14. CLOSURE CRITERIA
- 15. APPROVALS AND REVISION HISTORY

A.2 Scope Statement

Element	Content
Subject dimension	
Temporal boundary	
Geographic / jurisdictional boundary	
Activity boundary	
Inclusions	
Exclusions (each with a reason)	
Method boundaries	
Assumptions (each a registered risk)	

A.3 Objectives Register

Obj ID	Statement	Rationale	Success criterion	Completion test	Dependencies	Owner	Deadline	Priority	Status	Evidence produced
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A.4 Evidence Collection Plan Row

Row ID	Obj ID	Evidence sought	Category	Source/system	Collection method	Perishability	Legal basis / authorization	Documentation required at collection	Preservation procedure	Expected coverage boundary	Collector	Status
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A.5 Documentation Artifacts

Artifact	Format	Naming convention	Location	Cadence	Owner
Investigator notes				Continuous	
Search log				Per search	
Capture records				Per capture	
Evidence log				Per acquisition	
Custody log				Per event	
Decision log				Per decision	
Lead register				Continuous	
Communications record				Per exchange	
Production log				Per copy released	
Timeline				Continuous	

A.6 Chain of Custody Record

Entry #	Item ID	Event type	Date/time (TZ)	From (name, role)	To (name, role)	Purpose	Method	Hash before	Hash after	Result	Location after	Signatures	Notes
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A.7 Evidence Log

Item ID	Description	Category	Source / system	Acquired (date, time, method, by whom)	SHA-256	Size	Preservation location	Working copy location	Objective served	Coverage boundary
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A.8 Lead Register

Lead ID	Date raised	Source item/event	Description	In scope?	Priority	Owner	Status	Disposition and basis	Date closed
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A.9 Closure Memorandum

CASE NUMBER: _____ CLOSED: _____ BY: _____ APPROVED: _____

1. PURPOSE AND STRATEGIC GOAL
2. OBJECTIVES: status and basis for each
Complete · descoped (by whose decision) · blocked (by what)
3. WHAT WAS ACCOMPLISHED
4. WHAT WAS NOT, AND WHY
5. OPEN ITEMS AND HANDOFF
6. EXCEPTIONS
Custody exceptions · documentation gaps · unresolved requests
7. EVIDENCE RECONCILIATION
Items logged · items present · discrepancies · disposition
8. RETENTION AND DISPOSAL
What is kept · for how long · disposal date · authorized by
9. NOTES FOR A FUTURE INVESTIGATOR

A.10 Hashing Quick Reference

System	Single file	All files in a folder
Windows PowerShell	Get-FileHash -Algorithm SHA256 . \file	Get-FileHash -Algorithm SHA256 *
Windows Command Prompt	certutil -hashfile file SHA256	(loop, or use PowerShell)
macOS	shasum -a 256 file	shasum -a 256 *
Linux	sha256sum file	sha256sum *

Record the full 64-character value, the tool used, and the date and time.

Appendix B: The Planning and Custody Checklist

Plan approval (Section 1.9.2)

1. Authority recorded and adequate.
2. All four scope dimensions bounded.
3. Exclusions present and specific, each with a reason.
4. Method boundaries and a prohibited-activity statement present.
5. Assumptions registered.
6. Every objective measurable, with a success criterion and a completion test.
7. Every objective completable by a negative finding.
8. Priorities ranked by a stated method with a drop order.

9. Perishable evidence prioritized ahead of stable evidence.
10. Slow items scheduled first.
11. Every evidence row has a method, a legal basis, a documentation requirement, and a preservation procedure.
12. Coverage boundaries written in advance.
13. Documentation plan names artifacts, formats, naming, storage, and cadence.
14. Custody procedures cover acquisition, preservation, working copies, transfer, verification, production, and disposal.
15. Roles assigned, including who holds evidence, and control weaknesses named.
16. Review points scheduled.
17. Closure criteria stated.
18. Scope change procedure written, default stop.

Evidence handling, every item 19. Item ID assigned at acquisition, never reused. 20. Hash computed before anything else, full value recorded. 21. Source, method, time with zone, and collector recorded. 22. Original downloaded rather than re-created, where the item is a file. 23. Capture full-page with URL and timestamp, where the item is a page. 24. Query, filters, and result count recorded, where the item is an export. 25. Preservation copy created, verified, and never opened. 26. Working copy used for all analysis; annotations only on labelled copies. 27. Independent archive obtained where possible. 28. Coverage boundary recorded for every negative result.

Custody, ongoing 29. Every transfer hashed at both ends and recorded by both parties. 30. Every access logged, including your own. 31. Verification performed on schedule, all results recorded. 32. Production log maintained for every copy released. 33. No retroactive entries; late entries marked as late. 34. Breaks reported, extent determined, impact assessed, never reconstructed.

Closure 35. Every objective assessed with a basis. 36. Every lead dispositioned. 37. Inventory reconciled against the custody record. 38. Documentation passes the reproducibility test. 39. Exceptions disclosed in the closure memorandum and the report. 40. Retention applied and the disposal date calendared.

Appendix C: Where to Go Next

Skills to build after this chapter, in priority order:

1. **Write a one-page plan for every piece of research you do**, including personal research, for one month. The habit is the skill, and the one-page version is the one you will actually use.
2. **Hash and log something today**. Pick any file you care about, compute its SHA-256, record it, and verify it a week later. The mechanics take five minutes and remove the mystery

permanently.

3. **Run a reproducibility test on your own past work.** It is uncomfortable and it is the fastest way to find out which artifacts you are not producing.
4. **Learn your jurisdiction's public records process** by filing one request, tracking it, and recording the response including any exemptions applied.
5. **Practise writing exclusions.** Take any investigation you can imagine and write eight exclusions with reasons. It is harder than it sounds and it is the highest-value scope skill.
6. **Study a real custody standard** in whichever domain you intend to work: digital forensics, law enforcement property handling, regulated industry records, or litigation e-discovery. Conventions differ and the principles do not.
7. **Read a closure memorandum from a real investigation** if you can obtain one, and note what it discloses and what it omits.

Habits to start now:

- Write the scope before the search. Especially the exclusions.
 - Hash at acquisition. Always, and before anything else.
 - Never annotate an original.
 - Record what you did not find, with what you searched.
 - Give every lead a disposition, even "closed, unproductive, here is why."
 - When something breaks, report it. The disclosure is what makes you trustworthy.
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End of Chapter 6